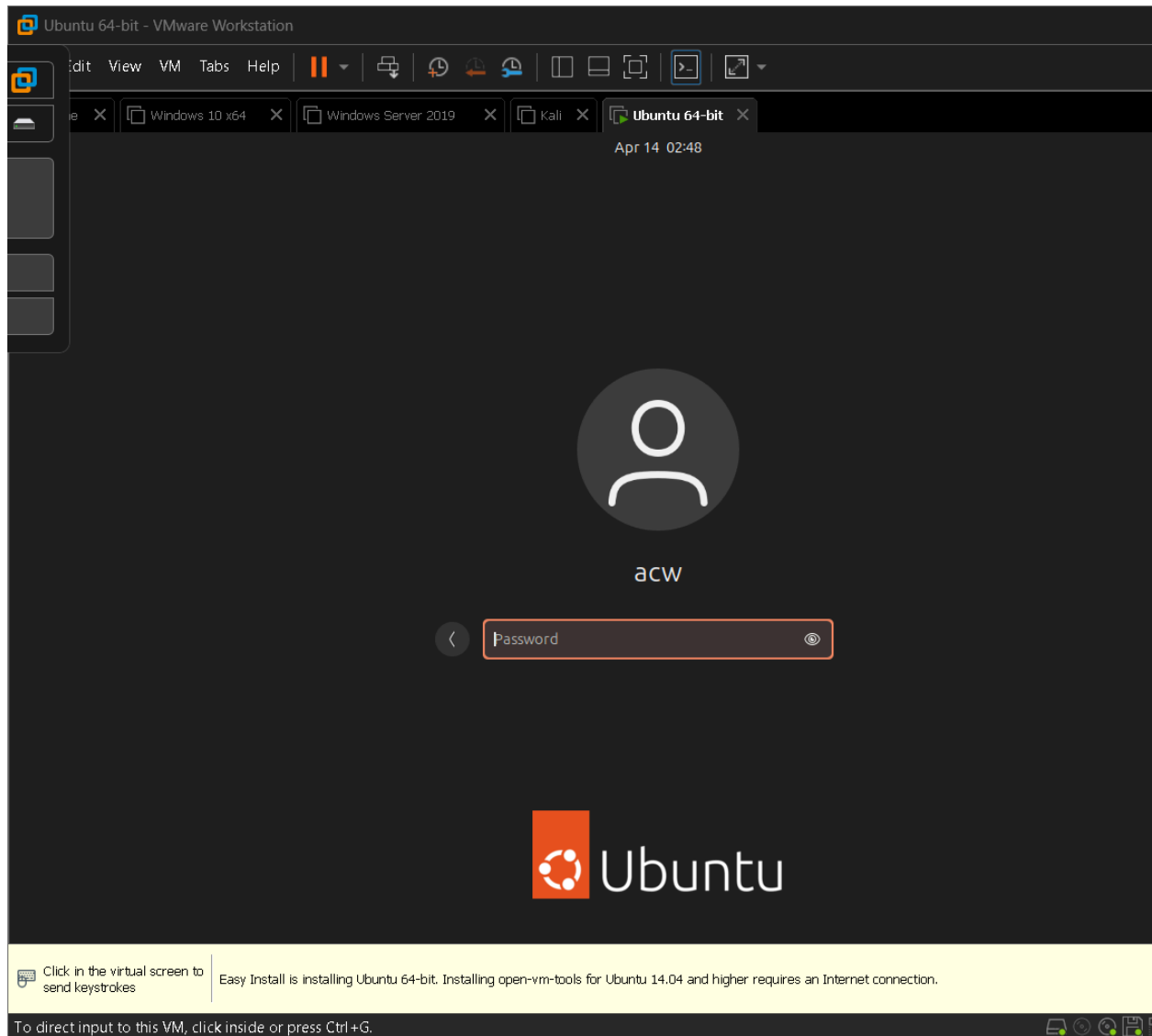
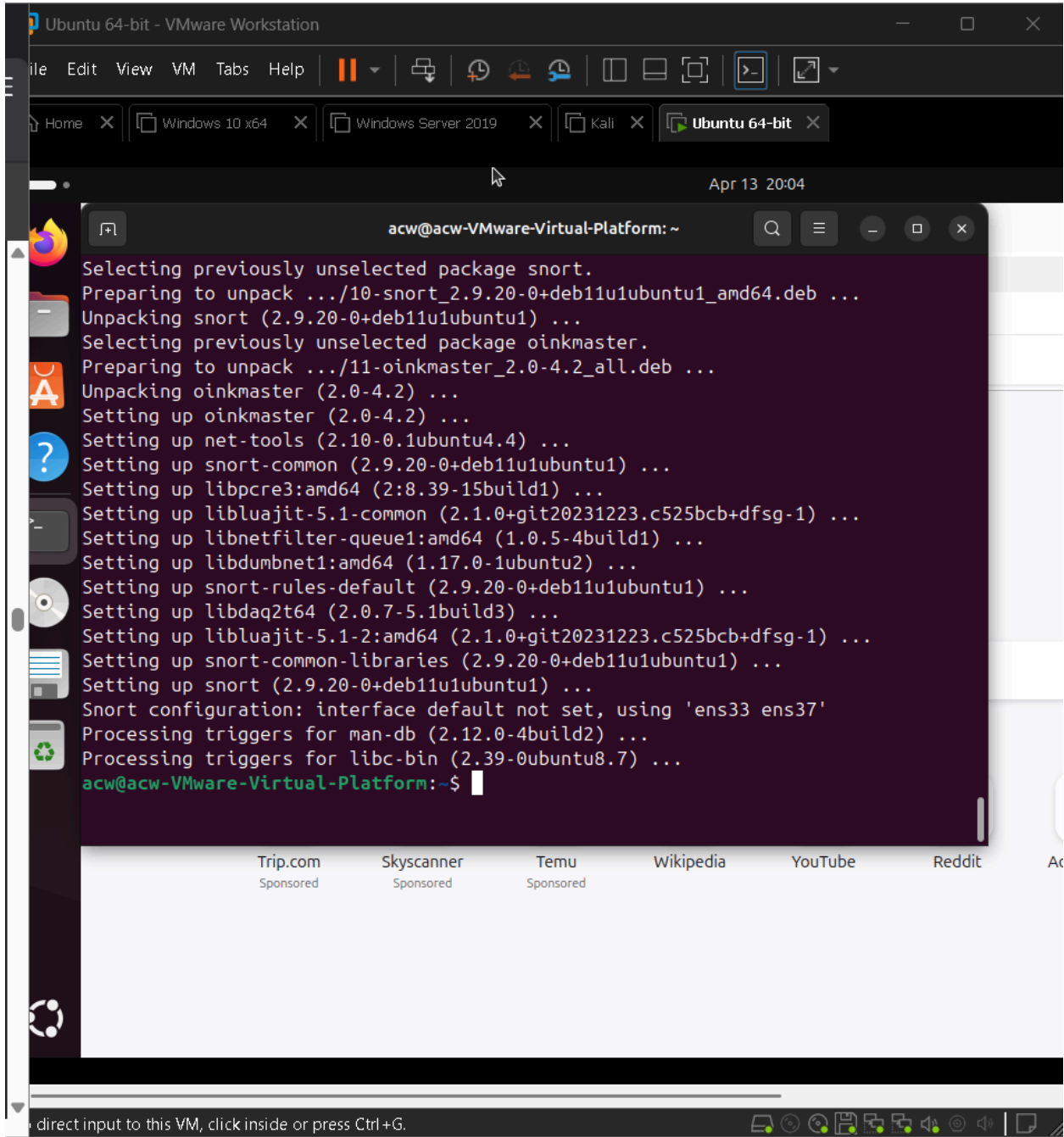


Lab Exercise 13.01: Installing Ubuntu and Snort

1v



2e

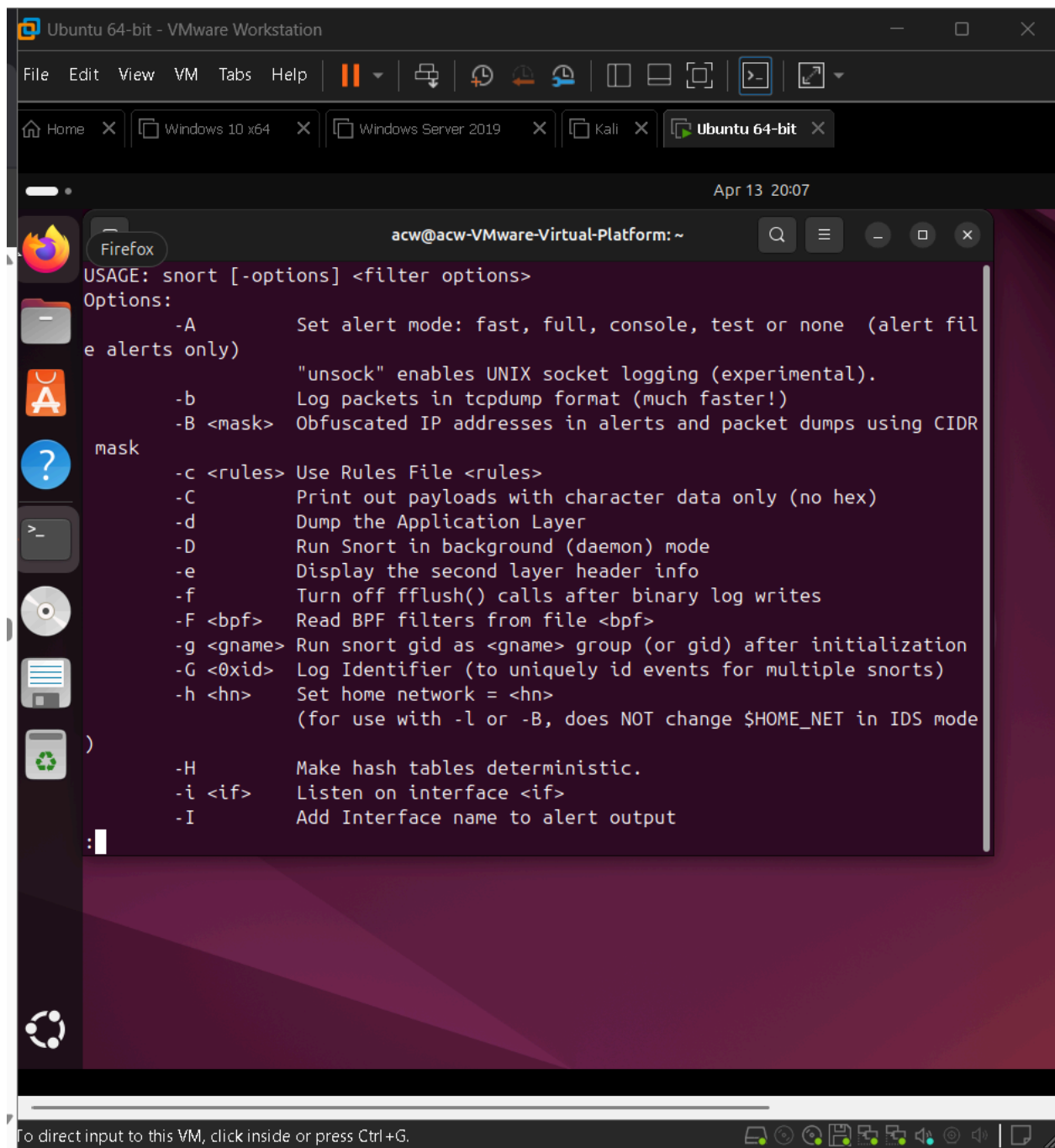


```
acw@acw-VMware-Virtual-Platform: ~  
Selecting previously unselected package snort.  
Preparing to unpack .../10-snort_2.9.20-0+deb11u1ubuntu1_amd64.deb ...  
Unpacking snort (2.9.20-0+deb11u1ubuntu1) ...  
Selecting previously unselected package oinkmaster.  
Preparing to unpack .../11-oinkmaster_2.0-4.2_all.deb ...  
Unpacking oinkmaster (2.0-4.2) ...  
Setting up oinkmaster (2.0-4.2) ...  
Setting up net-tools (2.10-0.1ubuntu4.4) ...  
Setting up snort-common (2.9.20-0+deb11u1ubuntu1) ...  
Setting up libpcap3:amd64 (2:8.39-15build1) ...  
Setting up liblua5.1-common (2.1.0+git20231223.c525bcb+dfsg-1) ...  
Setting up libnetfilter-queue1:amd64 (1.0.5-4build1) ...  
Setting up libdumbnet1:amd64 (1.17.0-1ubuntu2) ...  
Setting up snort-rules-default (2.9.20-0+deb11u1ubuntu1) ...  
Setting up libdaq2t64 (2.0.7-5.1build3) ...  
Setting up liblua5.1-2:amd64 (2.1.0+git20231223.c525bcb+dfsg-1) ...  
Setting up snort-common-libraries (2.9.20-0+deb11u1ubuntu1) ...  
Setting up snort (2.9.20-0+deb11u1ubuntu1) ...  
Snort configuration: interface default not set, using 'ens33 ens37'  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...  
acw@acw-VMware-Virtual-Platform:~$
```

Lab Exercise 13.02: Snort Sniffer Mode

1a-1c

1a



Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

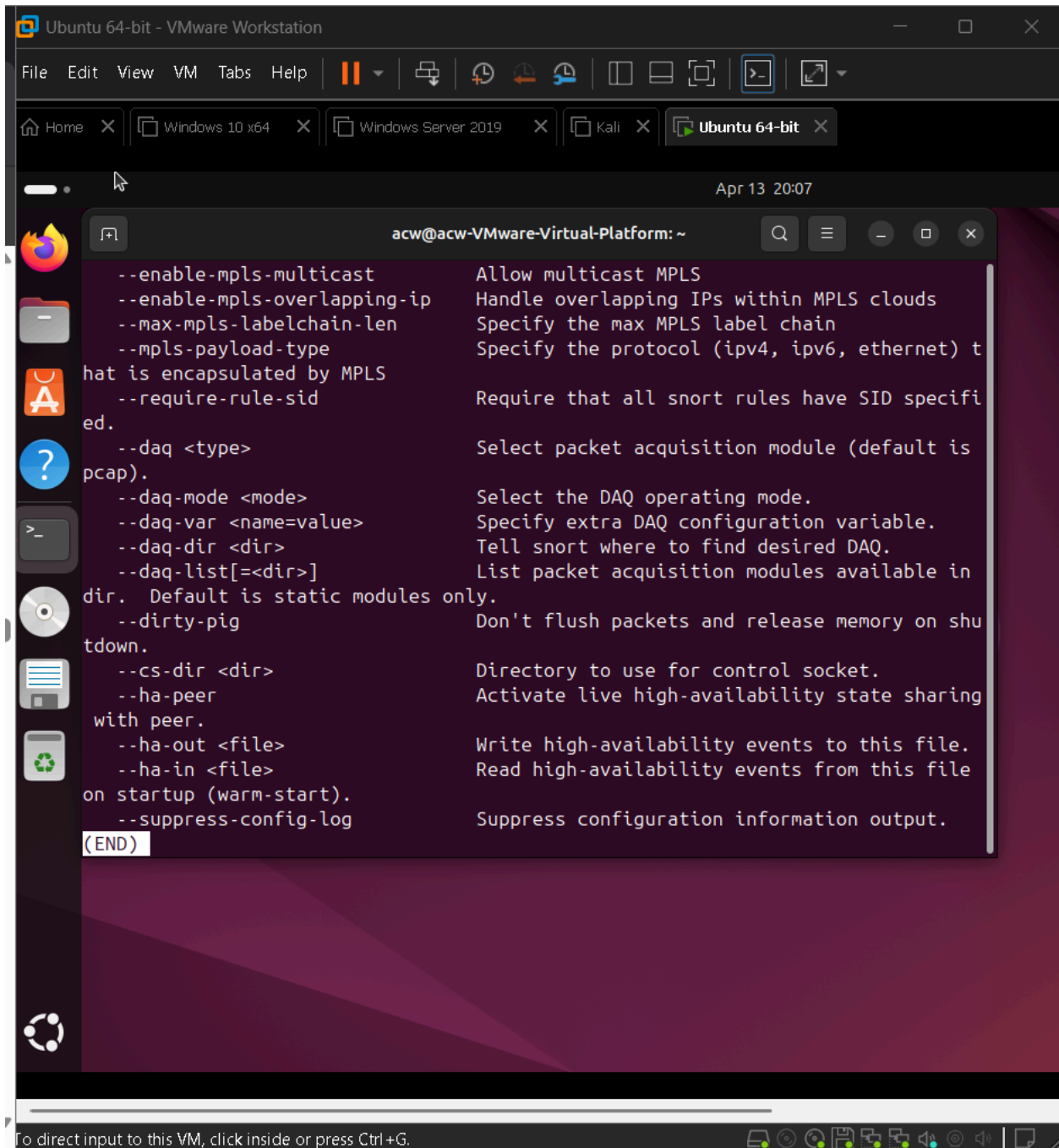
Home Windows 10 x64 Windows Server 2019 Kali Ubuntu 64-bit

Apr 13 20:07

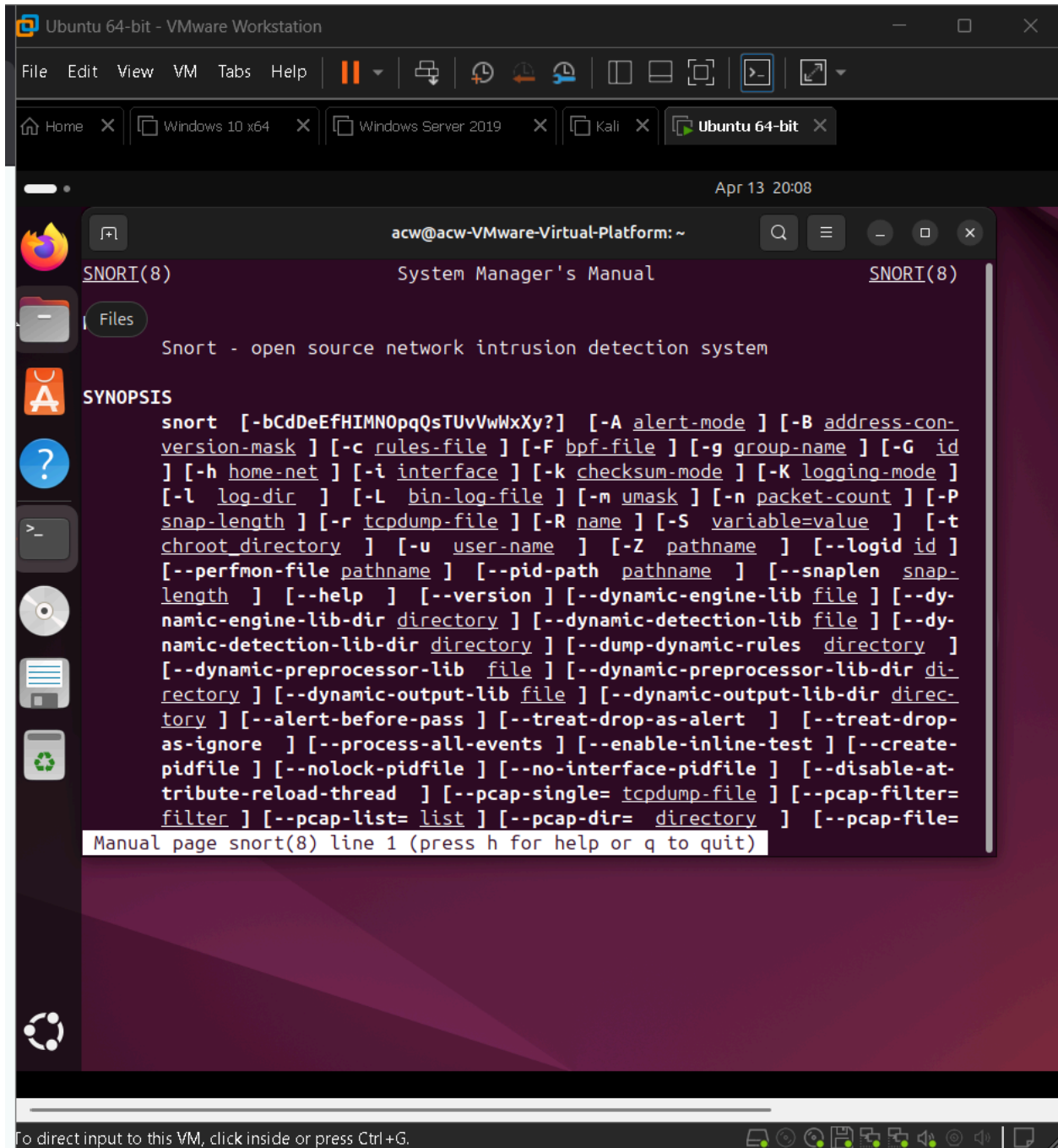
acw@acw-VMware-Virtual-Platform: ~

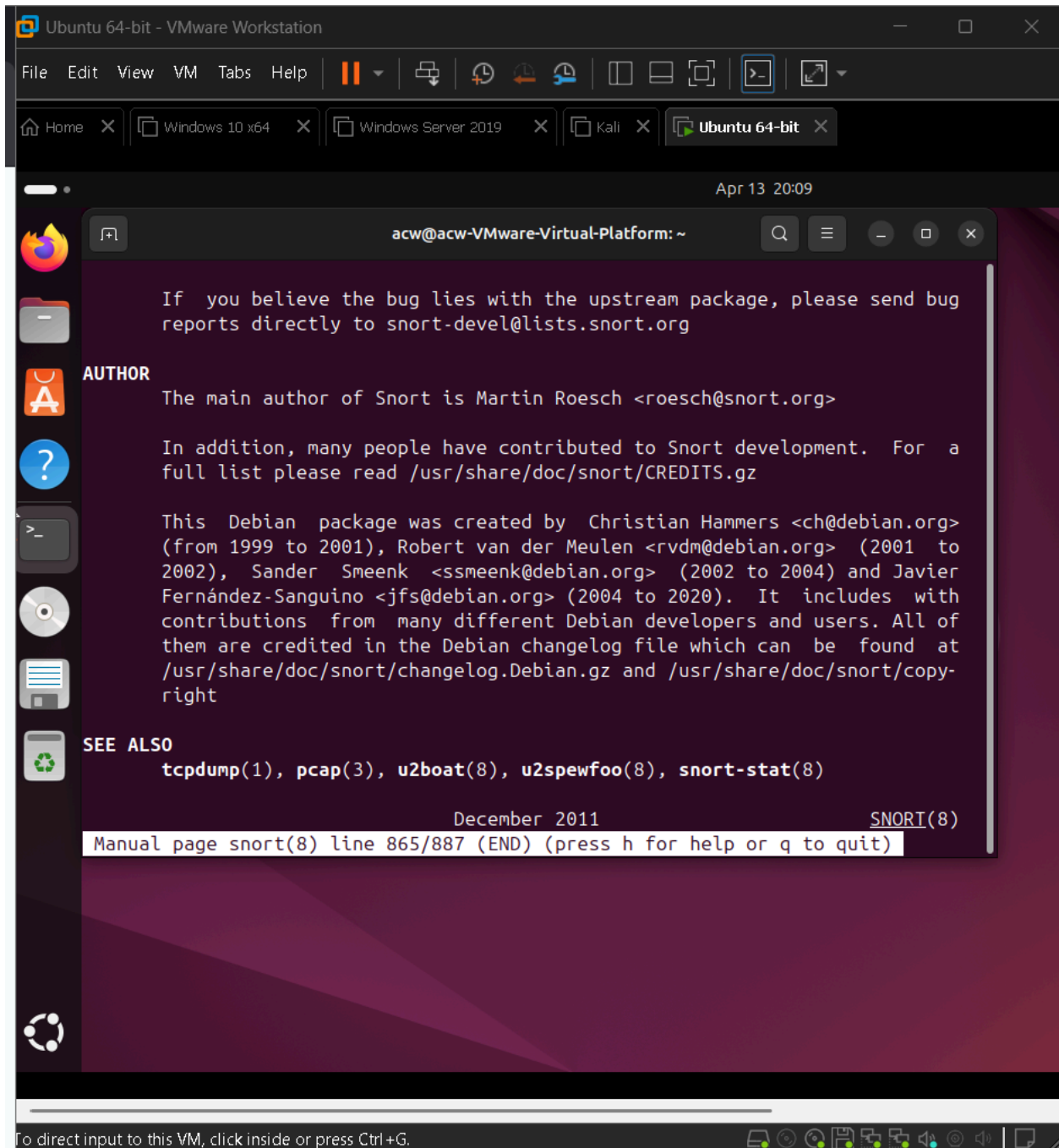
```
USAGE: snort [-options] <filter options>
Options:
  -A          Set alert mode: fast, full, console, test or none (alert file alerts only)
  -b          "unsock" enables UNIX socket logging (experimental).
              Log packets in tcpdump format (much faster!)
  -B <mask>   Obfuscated IP addresses in alerts and packet dumps using CIDR mask
  -c <rules>  Use Rules File <rules>
  -C          Print out payloads with character data only (no hex)
  -d          Dump the Application Layer
  -D          Run Snort in background (daemon) mode
  -e          Display the second layer header info
  -f          Turn off fflush() calls after binary log writes
  -F <bpf>    Read BPF filters from file <bpf>
  -g <gname>  Run snort gid as <gname> group (or gid) after initialization
  -G <0xid>   Log Identifier (to uniquely id events for multiple snorts)
  -h <hn>     Set home network = <hn>
              (for use with -l or -B, does NOT change $HOME_NET in IDS mode)
  -H          Make hash tables deterministic.
  -i <if>     Listen on interface <if>
  -I          Add Interface name to alert output
```

To direct input to this VM, click inside or press Ctrl+G.

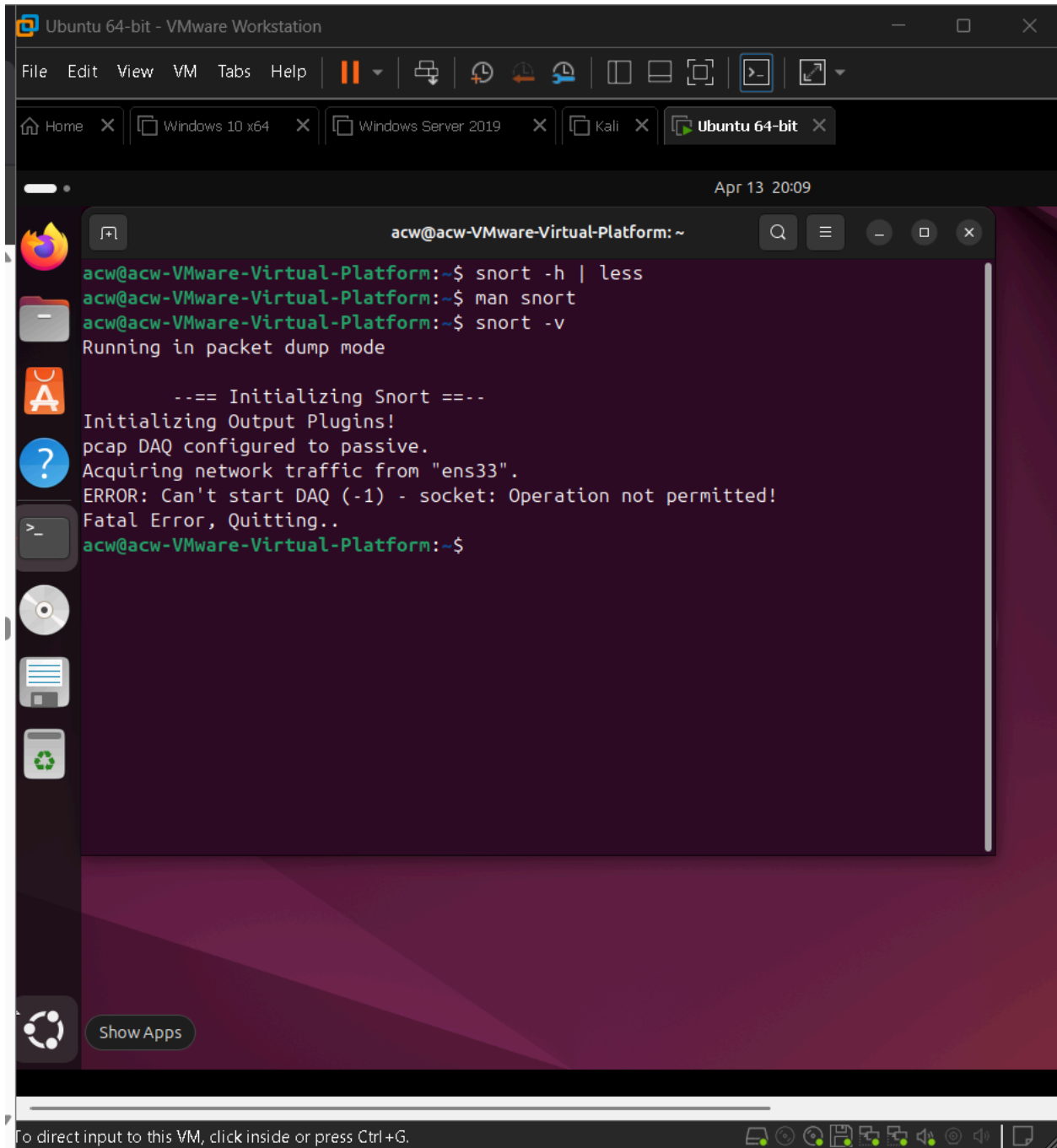


1b



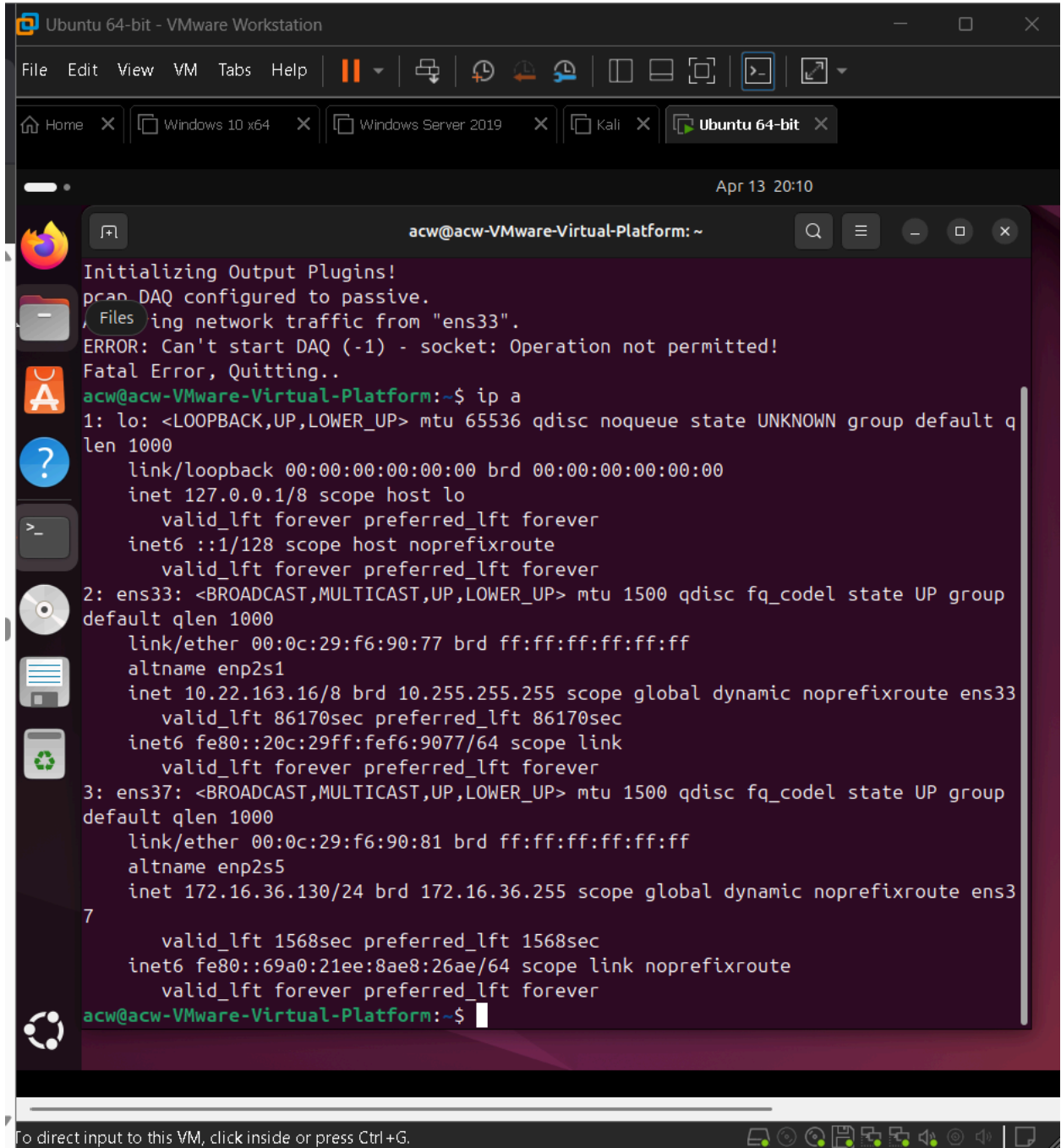


1c



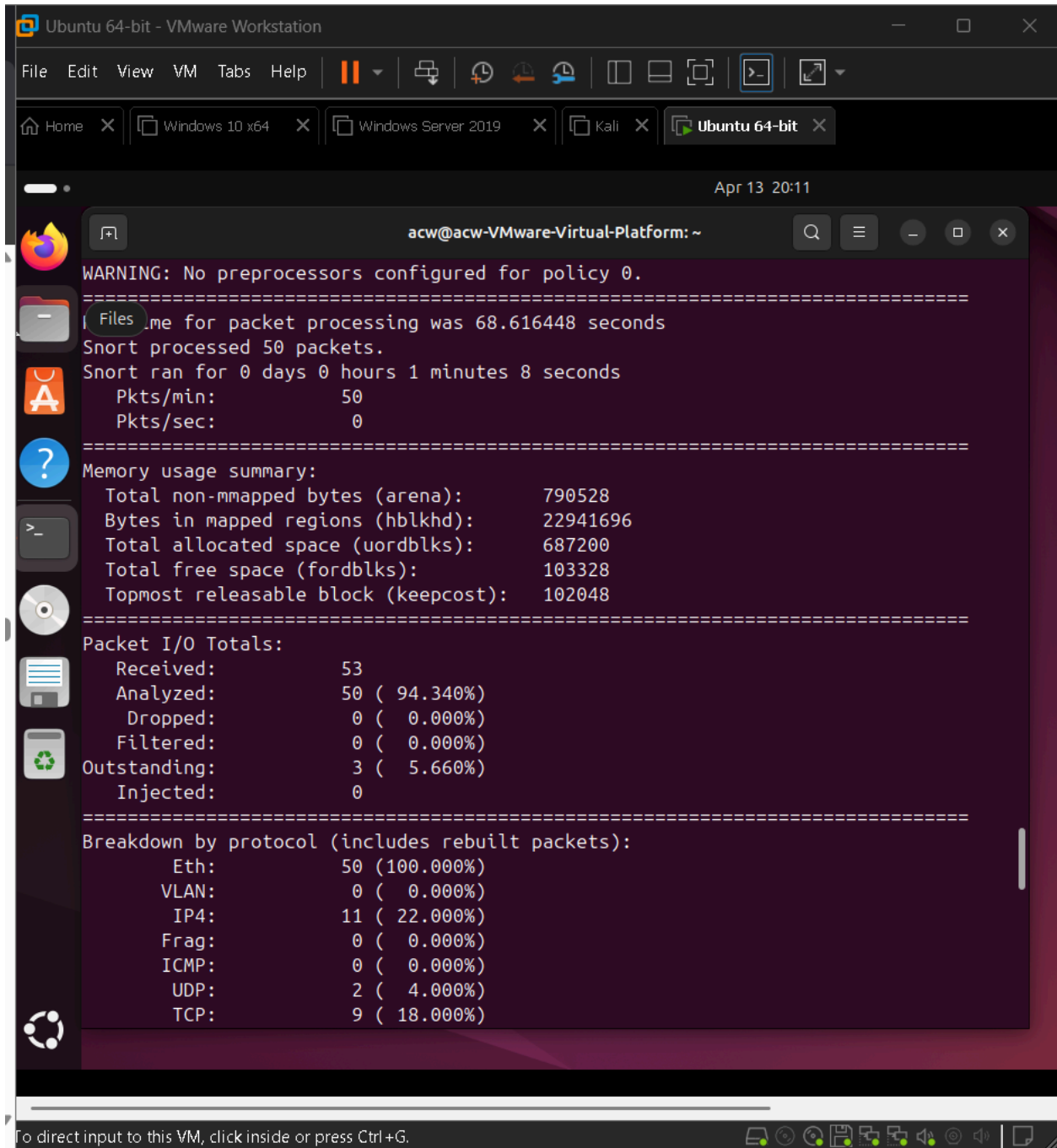
2a-2d

2a



```
acw@acw-VMware-Virtual-Platform: ~  
Initializing Output Plugins!  
pcan DAQ configured to passive.  
Files ing network traffic from "ens33".  
ERROR: Can't start DAQ (-1) - socket: Operation not permitted!  
Fatal Error, Quitting..  
acw@acw-VMware-Virtual-Platform:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default q  
len 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group  
default qlen 1000  
    link/ether 00:0c:29:f6:90:77 brd ff:ff:ff:ff:ff:ff  
    altname enp2s1  
    inet 10.22.163.16/8 brd 10.255.255.255 scope global dynamic noprefixroute ens33  
        valid_lft 86170sec preferred_lft 86170sec  
    inet6 fe80::20c:29ff:fe6:9077/64 scope link  
        valid_lft forever preferred_lft forever  
3: ens37: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group  
default qlen 1000  
    link/ether 00:0c:29:f6:90:81 brd ff:ff:ff:ff:ff:ff  
    altname enp2s5  
    inet 172.16.36.130/24 brd 172.16.36.255 scope global dynamic noprefixroute ens3  
7  
        valid_lft 1568sec preferred_lft 1568sec  
    inet6 fe80::69a0:21ee:8ae8:26ae/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
acw@acw-VMware-Virtual-Platform:~$
```

2b



Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

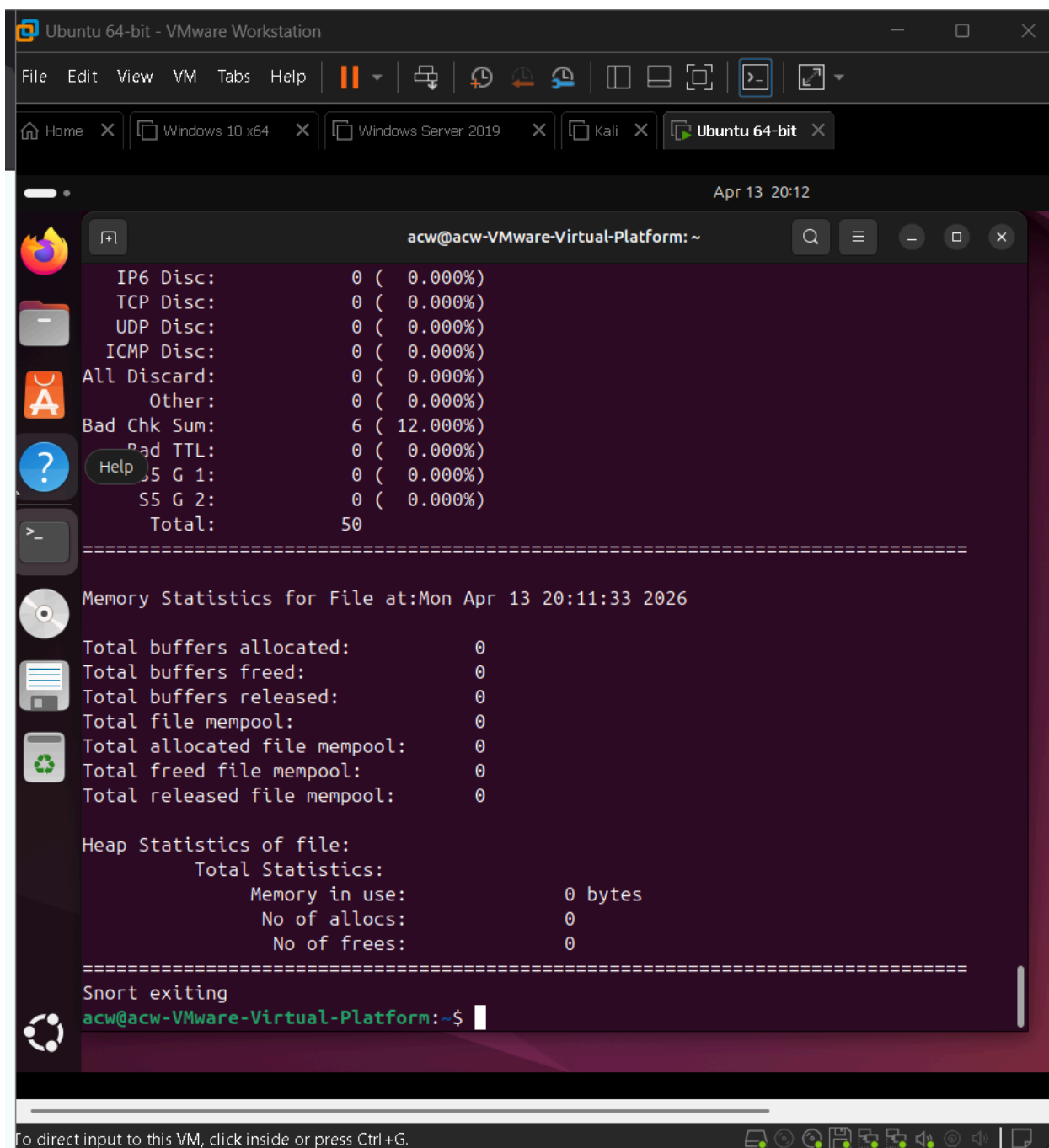
Home X Windows 10 x64 X Windows Server 2019 X Kali X Ubuntu 64-bit X

Apr 13 20:11

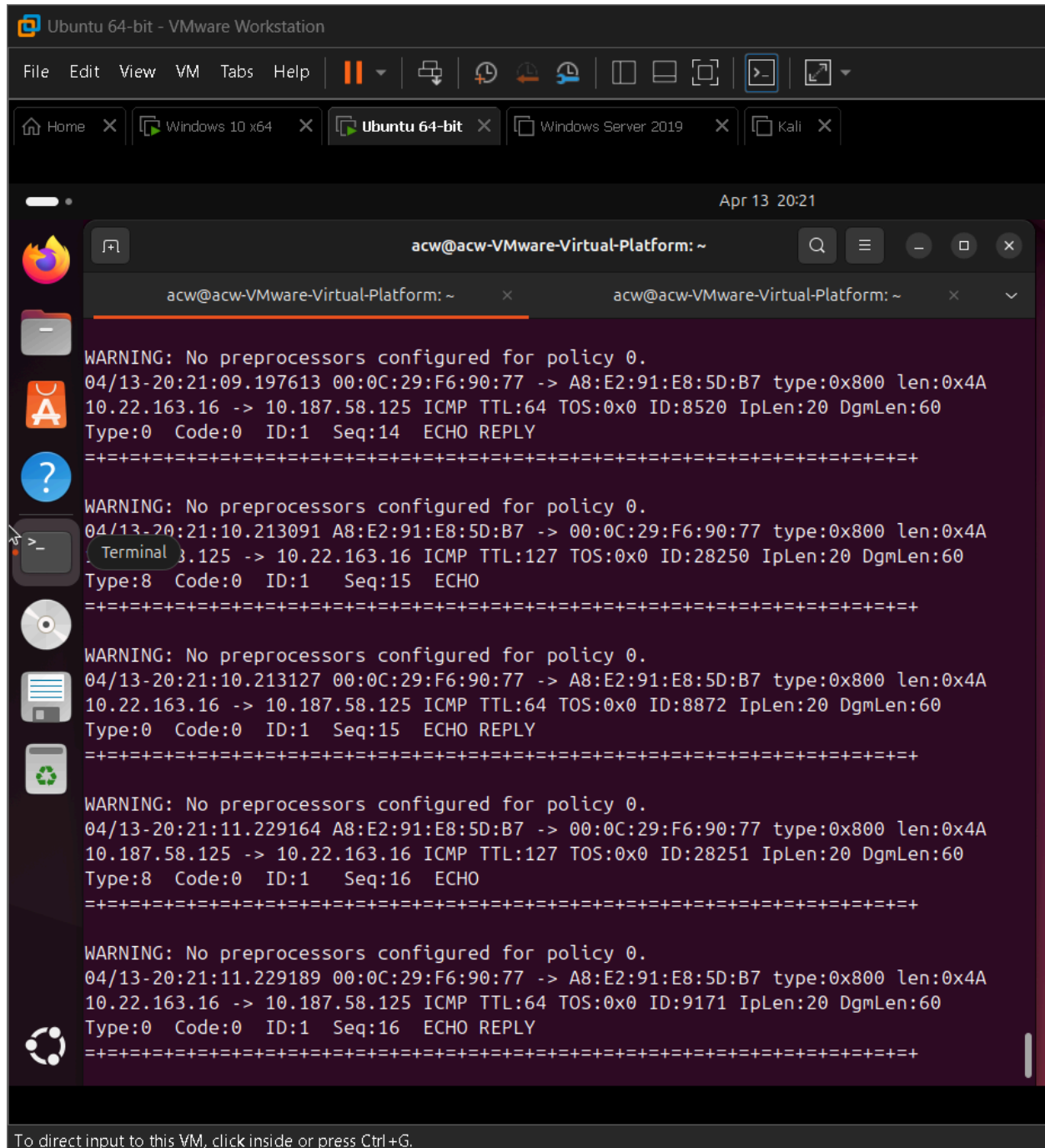
acw@acw-VMware-Virtual-Platform: ~

```
WARNING: No preprocessors configured for policy 0.
=====
Time for packet processing was 68.616448 seconds
Snort processed 50 packets.
Snort ran for 0 days 0 hours 1 minutes 8 seconds
  Pkts/min:      50
  Pkts/sec:       0
=====
Memory usage summary:
  Total non-mmapped bytes (arena):      790528
  Bytes in mapped regions (hblkhd):     22941696
  Total allocated space (uordblks):     687200
  Total free space (fordblks):          103328
  Topmost releasable block (keepcost):  102048
=====
Packet I/O Totals:
  Received:      53
  Analyzed:      50 ( 94.340%)
  Dropped:       0 (  0.000%)
  Filtered:      0 (  0.000%)
  Outstanding:   3 (  5.660%)
  Injected:      0
=====
Breakdown by protocol (includes rebuilt packets):
  Eth:           50 (100.000%)
  VLAN:          0 (  0.000%)
  IP4:           11 ( 22.000%)
  Frag:          0 (  0.000%)
  ICMP:          0 (  0.000%)
  UDP:           2 (  4.000%)
  TCP:           9 ( 18.000%)
```

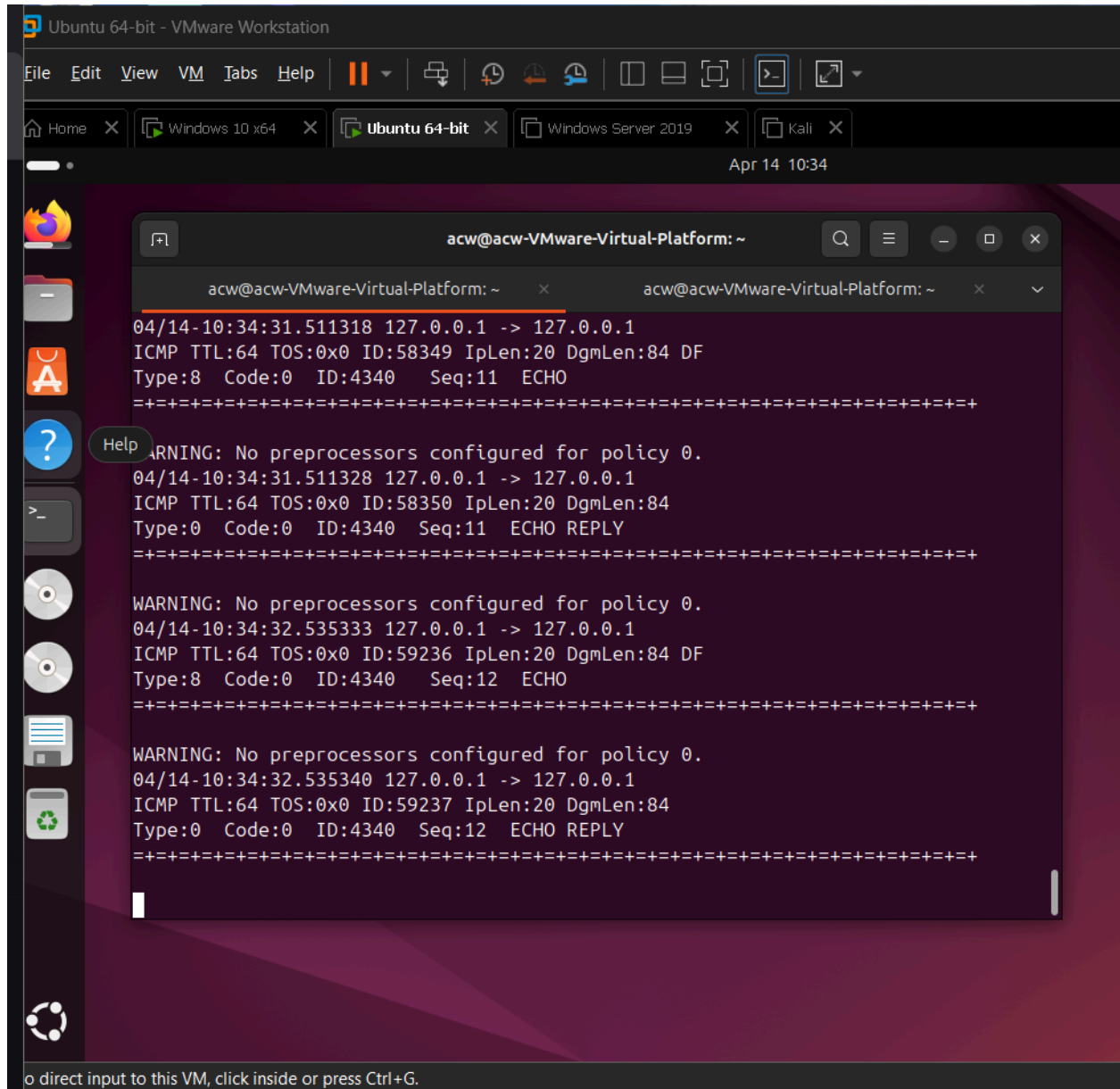
To direct input to this VM, click inside or press Ctrl+G.

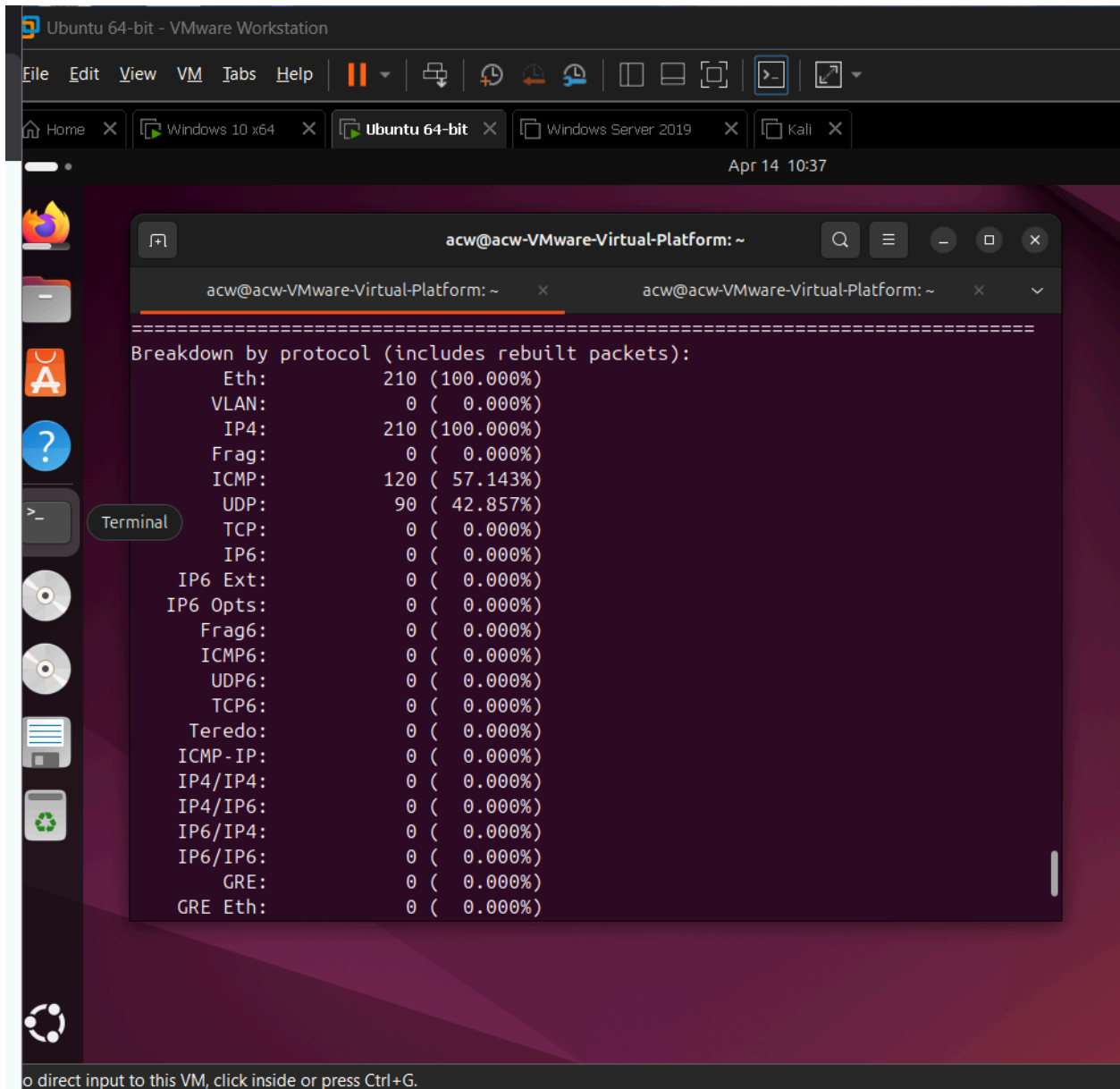


2c

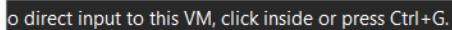


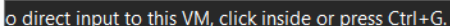
2d



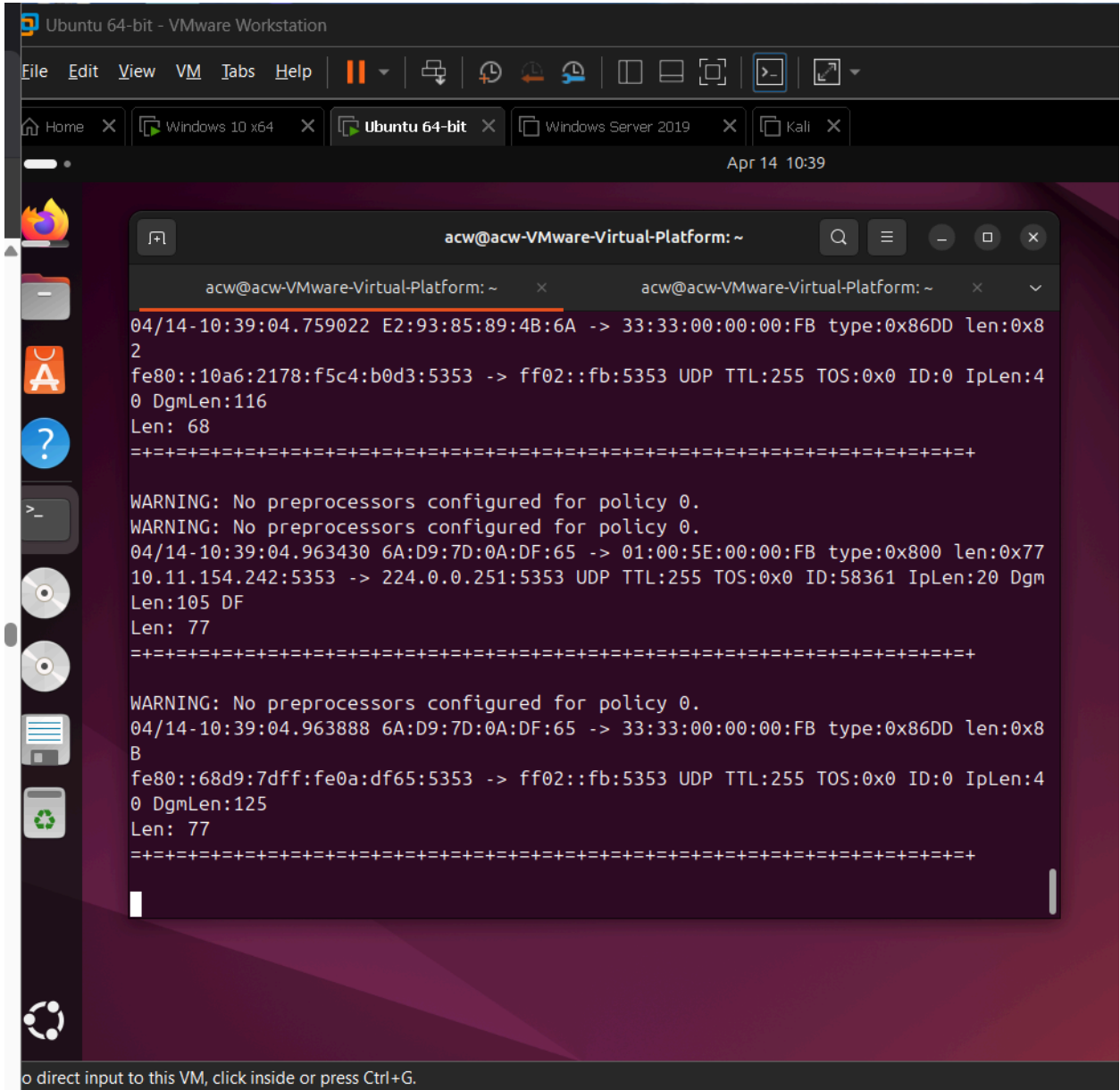


3a

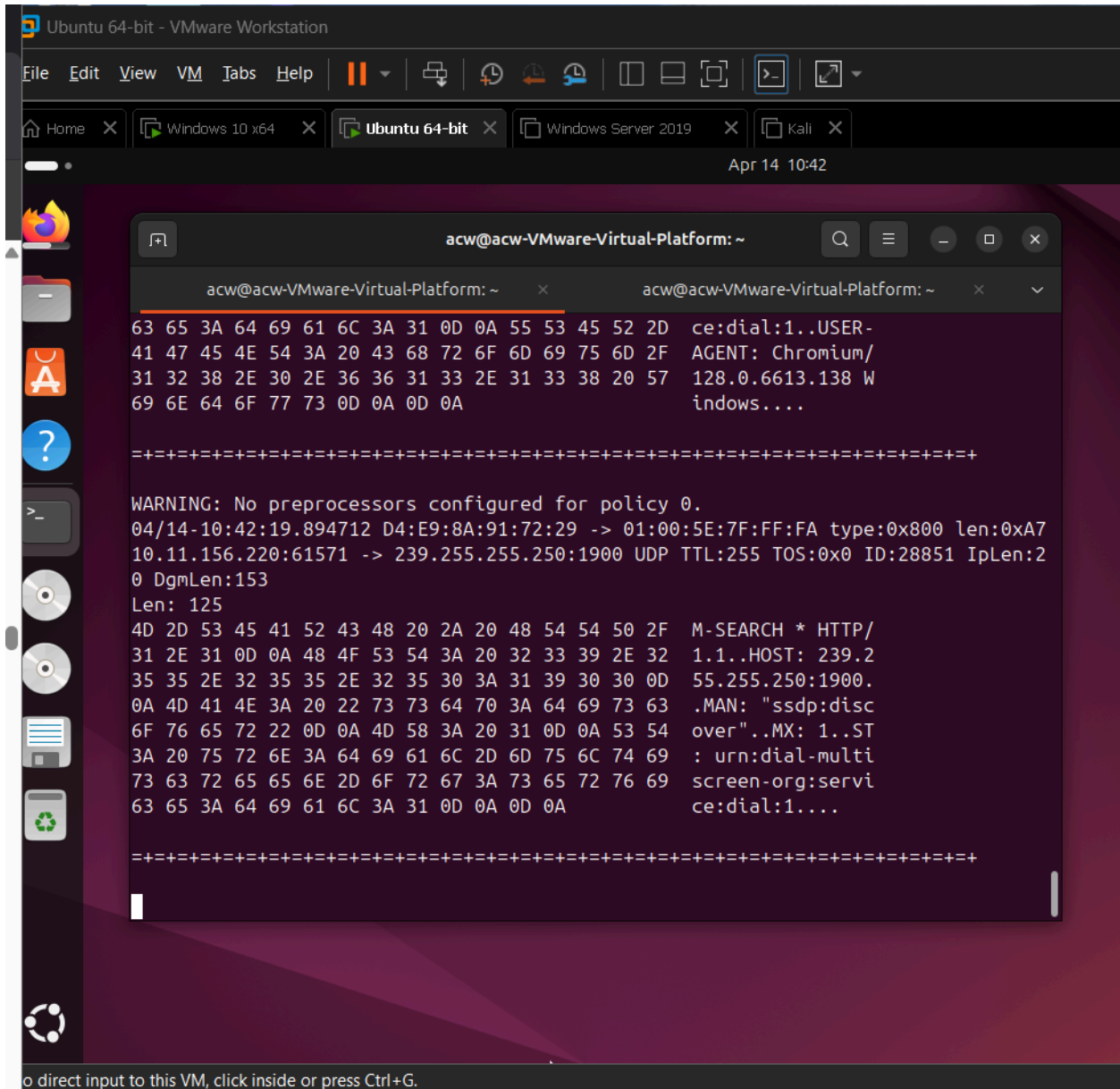




3b



3c



Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

Home x Windows 10 x64 x Ubuntu 64-bit x Windows Server 2019 x Kali x

Apr 14 10:42

acw@acw-VMware-Virtual-Platform: ~

```
acw@acw-VMware-Virtual-Platform: ~ x acw@acw-VMware-Virtual-Platform: ~ x v
63 65 3A 64 69 61 6C 3A 31 0D 0A 55 53 45 52 2D ce:dial:1..USER-
41 47 45 4E 54 3A 20 43 68 72 6F 6D 69 75 6D 2F AGENT: Chromium/
31 32 38 2E 30 2E 36 36 31 33 2E 31 33 38 20 57 128.0.6613.138 W
69 6E 64 6F 77 73 0D 0A 0D 0A indows....

==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+

WARNING: No preprocessors configured for policy 0.
04/14-10:42:19.894712 D4:E9:8A:91:72:29 -> 01:00:5E:7F:FF:FA type:0x800 len:0xA7
10.11.156.220:61571 -> 239.255.255.250:1900 UDP TTL:255 TOS:0x0 ID:28851 IpLen:2
0 DgmLen:153
Len: 125
4D 2D 53 45 41 52 43 48 20 2A 20 48 54 54 50 2F M-SEARCH * HTTP/
31 2E 31 0D 0A 48 4F 53 54 3A 20 32 33 39 2E 32 1.1..HOST: 239.2
35 35 2E 32 35 35 2E 32 35 30 3A 31 39 30 30 0D 55.255.250:1900.
0A 4D 41 4E 3A 20 22 73 73 64 70 3A 64 69 73 63 .MAN: "ssdp:disc
6F 76 65 72 22 0D 0A 4D 58 3A 20 31 0D 0A 53 54 over"..MX: 1..ST
3A 20 75 72 6E 3A 64 69 61 6C 2D 6D 75 6C 74 69 : urn:dial-multi
73 63 72 65 65 6E 2D 6F 72 67 3A 73 65 72 76 69 screen-org:servi
63 65 3A 64 69 61 6C 3A 31 0D 0A 0D 0A ce:dial:1....

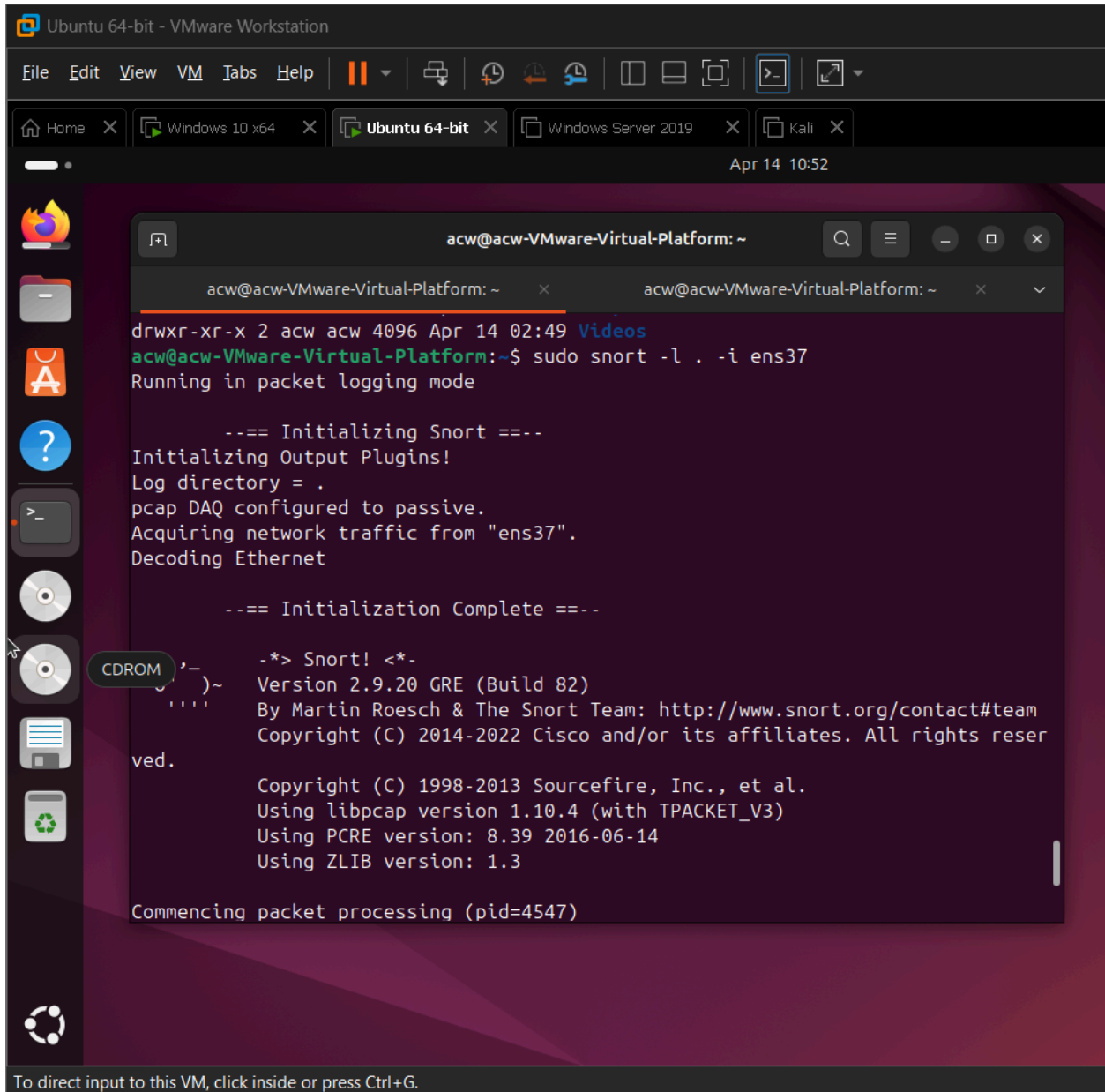
==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+==+
```

to direct input to this VM, click inside or press Ctrl+G.

Lab Exercise 13.03: Snort Packet Logger Mode

1a, 1b

1a

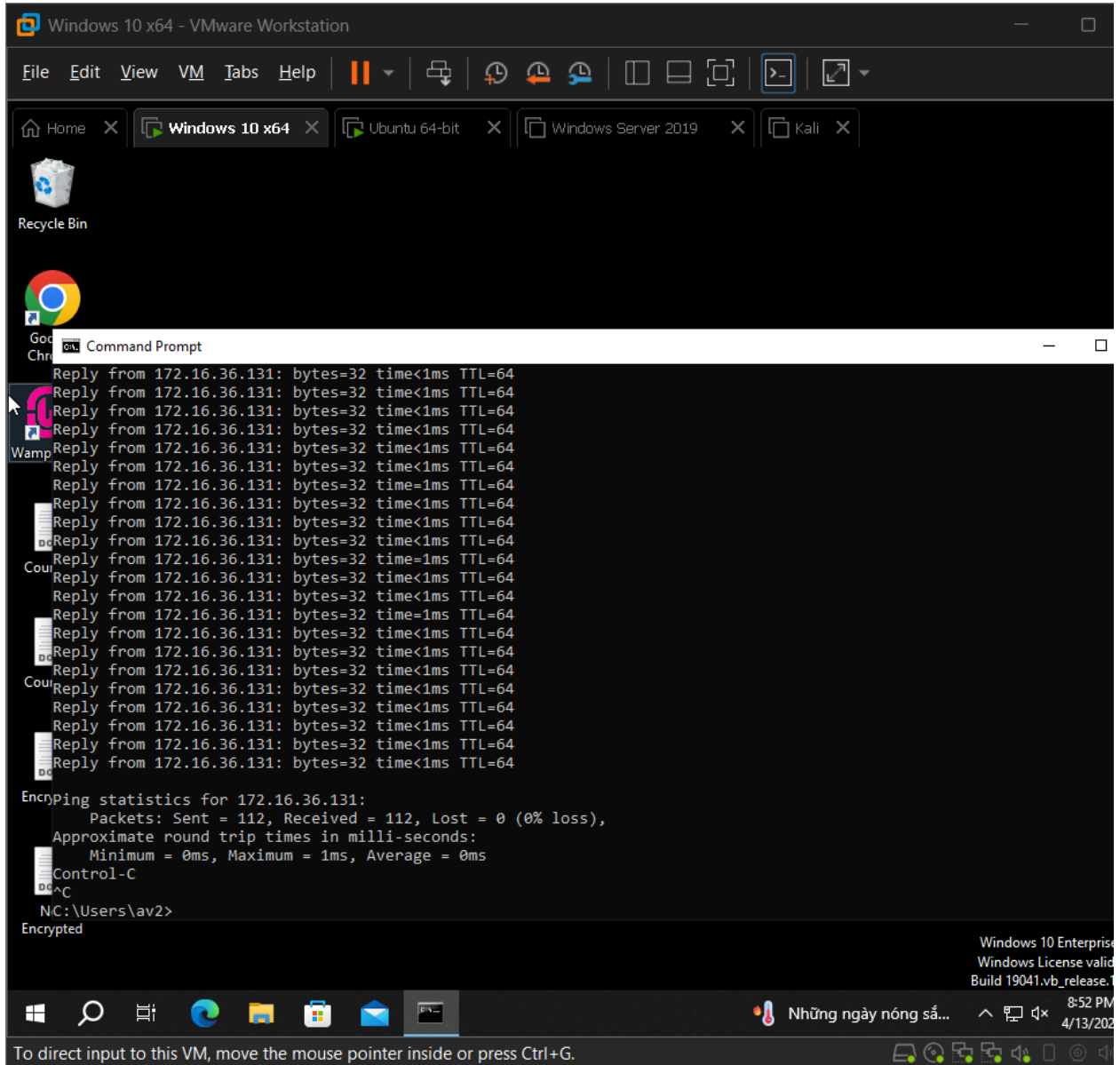


The screenshot shows a VMware Workstation window titled "Ubuntu 64-bit - VMware Workstation". The interface includes a menu bar (File, Edit, View, VM, Tabs, Help), a toolbar with icons for power, undo, redo, and other VM controls, and a tab bar showing four virtual machines: Home, Windows 10 x64, Ubuntu 64-bit (selected), Windows Server 2019, and Kali. The main display area shows the Ubuntu desktop with a dark purple background and a sidebar of application icons. A terminal window is open in the foreground, displaying the following text:

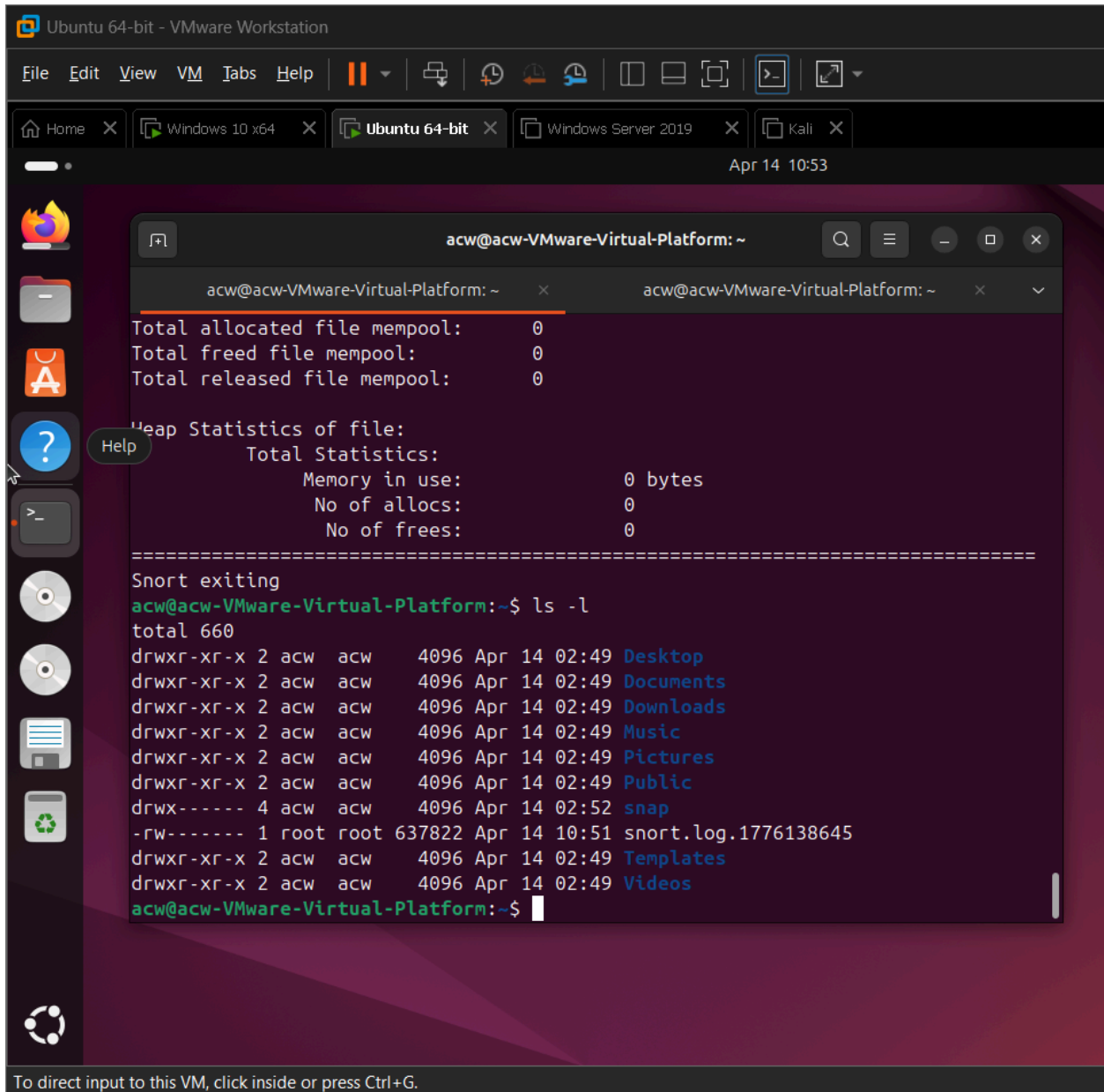
```
acw@acw-VMware-Virtual-Platform: ~  
drwxr-xr-x 2 acw acw 4096 Apr 14 02:49 Videos  
acw@acw-VMware-Virtual-Platform:~$ sudo snort -l . -i ens37  
Running in packet logging mode  
  
--== Initializing Snort ==--  
Initializing Output Plugins!  
Log directory = .  
pcap DAQ configured to passive.  
Acquiring network traffic from "ens37".  
Decoding Ethernet  
  
--== Initialization Complete ==--  
  
-*> Snort! <*-  
Version 2.9.20 GRE (Build 82)  
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team  
Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.  
Copyright (C) 1998-2013 Sourcefire, Inc., et al.  
Using libpcap version 1.10.4 (with TPACKET_V3)  
Using PCRE version: 8.39 2016-06-14  
Using ZLIB version: 1.3  
  
Commencing packet processing (pid=4547)
```

At the bottom of the VMware window, a status bar reads: "To direct input to this VM, click inside or press Ctrl+G."

1b



2a



Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

Home Windows 10 x64 Ubuntu 64-bit Windows Server 2019 Kali

Apr 14 10:53

```
acw@acw-VMware-Virtual-Platform: ~  
acw@acw-VMware-Virtual-Platform: ~  
Total allocated file mempool: 0  
Total freed file mempool: 0  
Total released file mempool: 0  
Heap Statistics of file:  
Total Statistics:  
Memory in use: 0 bytes  
No of allocs: 0  
No of frees: 0  
=====
```

Snort exiting

```
acw@acw-VMware-Virtual-Platform:~$ ls -l  
total 660  
drwxr-xr-x 2 acw acw 4096 Apr 14 02:49 Desktop  
drwxr-xr-x 2 acw acw 4096 Apr 14 02:49 Documents  
drwxr-xr-x 2 acw acw 4096 Apr 14 02:49 Downloads  
drwxr-xr-x 2 acw acw 4096 Apr 14 02:49 Music  
drwxr-xr-x 2 acw acw 4096 Apr 14 02:49 Pictures  
drwxr-xr-x 2 acw acw 4096 Apr 14 02:49 Public  
drwx----- 4 acw acw 4096 Apr 14 02:52 snap  
-rw----- 1 root root 637822 Apr 14 10:51 snort.log.1776138645  
drwxr-xr-x 2 acw acw 4096 Apr 14 02:49 Templates  
drwxr-xr-x 2 acw acw 4096 Apr 14 02:49 Videos  
acw@acw-VMware-Virtual-Platform:~$
```

To direct input to this VM, click inside or press Ctrl+G.

2b

Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

Home Windows 10 x64 Ubuntu 64-bit Windows Server 2019 Kali

Apr 14 10:54

snort.log.1776138645

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
24	0.836777	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
25	0.836815	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
48	1.851604	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
49	1.851625	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
60	2.868614	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
61	2.868633	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
89	3.883052	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
90	3.883076	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
101	4.898985	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
102	4.899011	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
150	5.914908	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
151	5.914930	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
165	6.930139	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
166	6.930161	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
167	7.945738	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
168	7.945762	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply

Frame 24: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
 Ethernet II, Src: VMware_a8:b4:29 (00:0c:29:a8:b4:29), Dst: 08:00:27:87:02:7a (08:00:27:87:02:7a)
 Internet Protocol Version 4, Src: 172.16.36.128, Destination: 172.16.36.131
 Internet Control Message Protocol

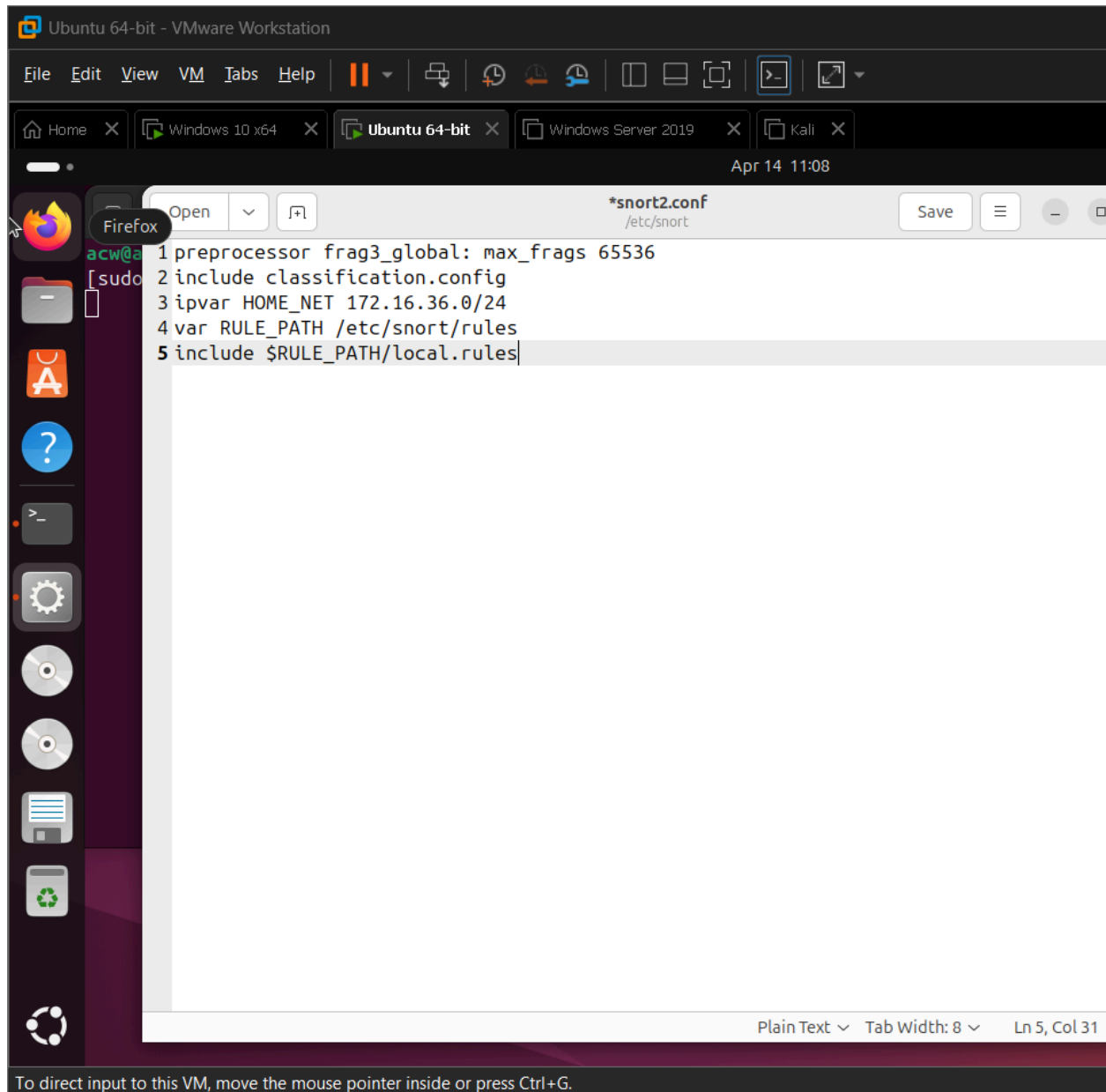
0000 00 0c 29 87 02 7a 00 0c 29 a8 b4 29 08
 0010 00 3c d5 5b 00 00 80 01 c4 41 ac 10 24
 0020 24 83 08 00 4c 75 00 01 00 e6 61 62 63
 0030 67 68 69 6a 6b 6c 6d 6e 6f 70 71 72 73
 0040 77 61 62 63 64 65 66 67 68 69

Internet Control Message Protocol: Protocol Packets: 668 - Displayed: 52 (7.8%) Profile: D

To direct input to this VM, click inside or press Ctrl+G.

Lab Exercise 13.04: Snort Network Intrusion Detection System Mode

1a–1k



Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

Home Windows 10 x64 Ubuntu 64-bit Windows Server 2019 Kali

Apr 14 11:09

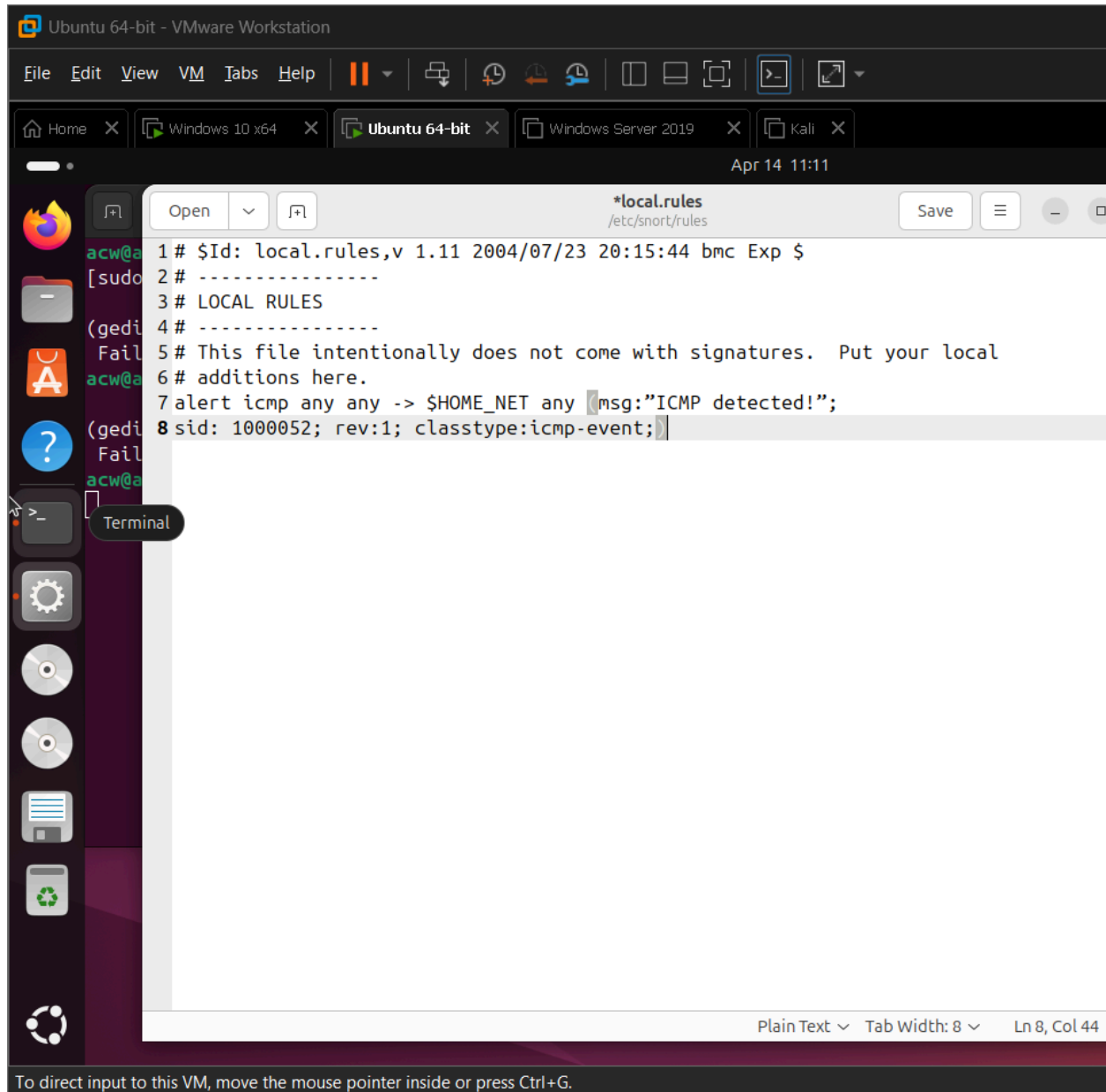
Open classification.config /etc/snort Save

```
1 # $Id$
2 # The following includes information for prioritizing rules
3 #
4 # Each classification includes a shortname, a description, and a default
5 # priority for that classification.
6 #
7 # This allows alerts to be classified and prioritized. You can specify
8 # what priority each classification has. Any rule can override the default
9 # priority for that rule.
10 #
11 # Here are a few example rules:
12 #
13 #   alert TCP any any -> any 80 (msg: "EXPLOIT ntpdx overflow";
14 #       dsize: > 128; classtype:attempted-admin; priority:10;
15 #
16 #   alert TCP any any -> any 25 (msg:"SMTP expn root"; flags:A+; \
17 #       content:"expn root"; nocase; classtype:attempted-recon;)
18 #
19 # The first rule will set its type to "attempted-admin" and override
20 # the default priority for that type to 10.
21 #
22 # The second rule set its type to "attempted-recon" and set its
23 # priority to the default for that type.
24 #
25
26 #
27 # config classification:shortname,short description,priority
28 #
29
30 config classification: not-suspicious,Not Suspicious Traffic,3
31 config classification: unknown,Unknown Traffic,3
```

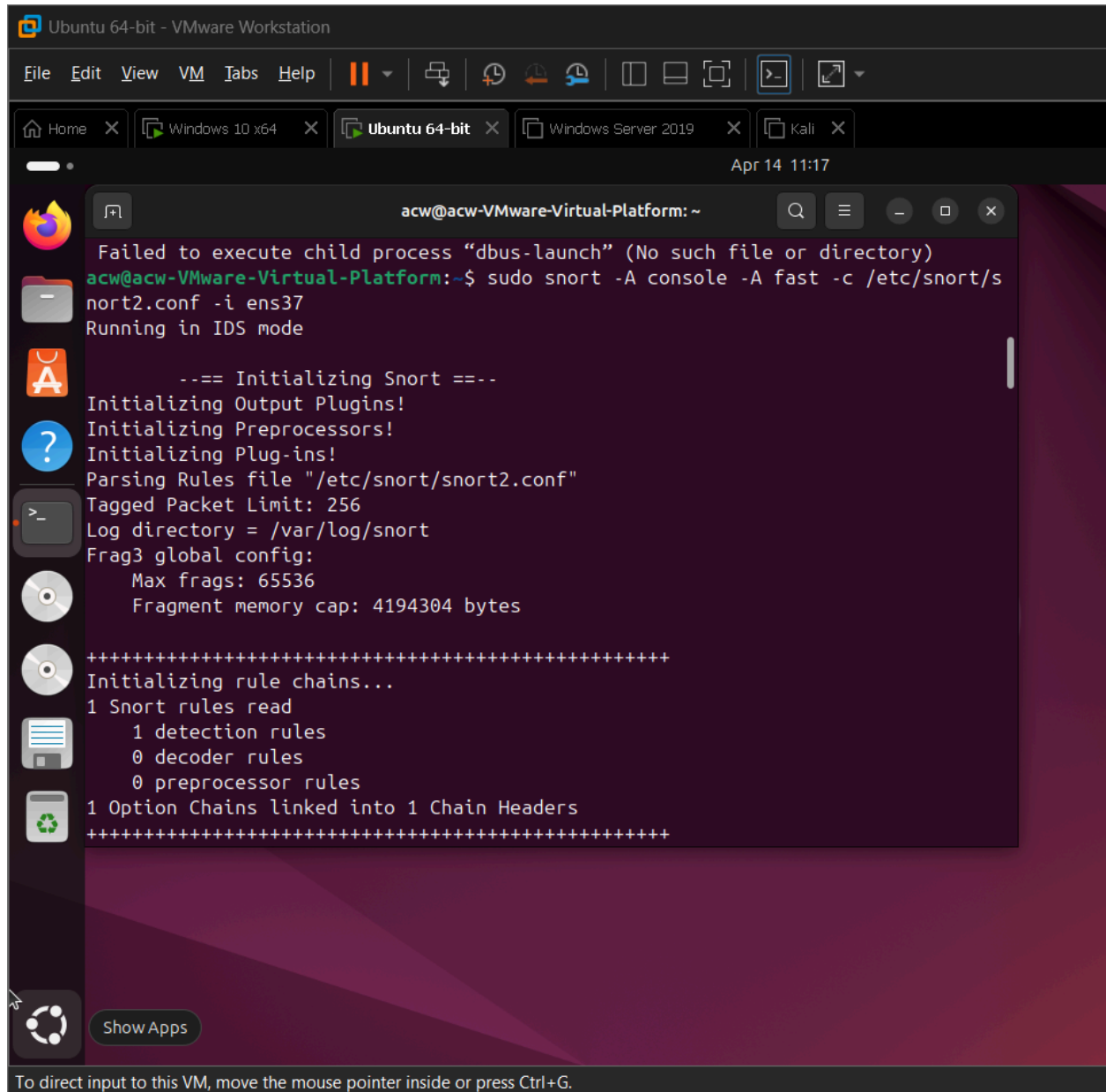
Plain Text Tab Width: 8 Ln 1, Col 1

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

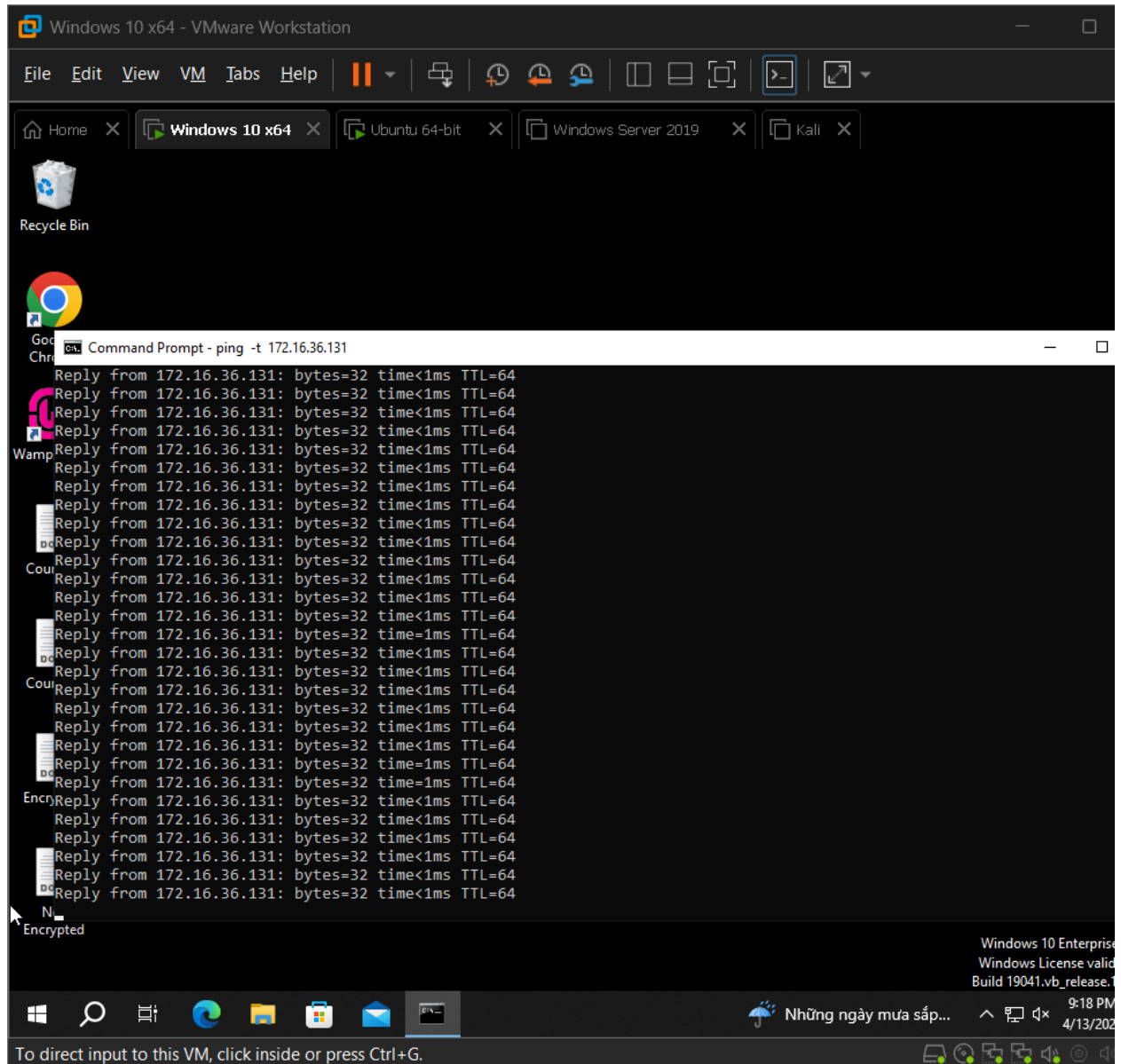
1b

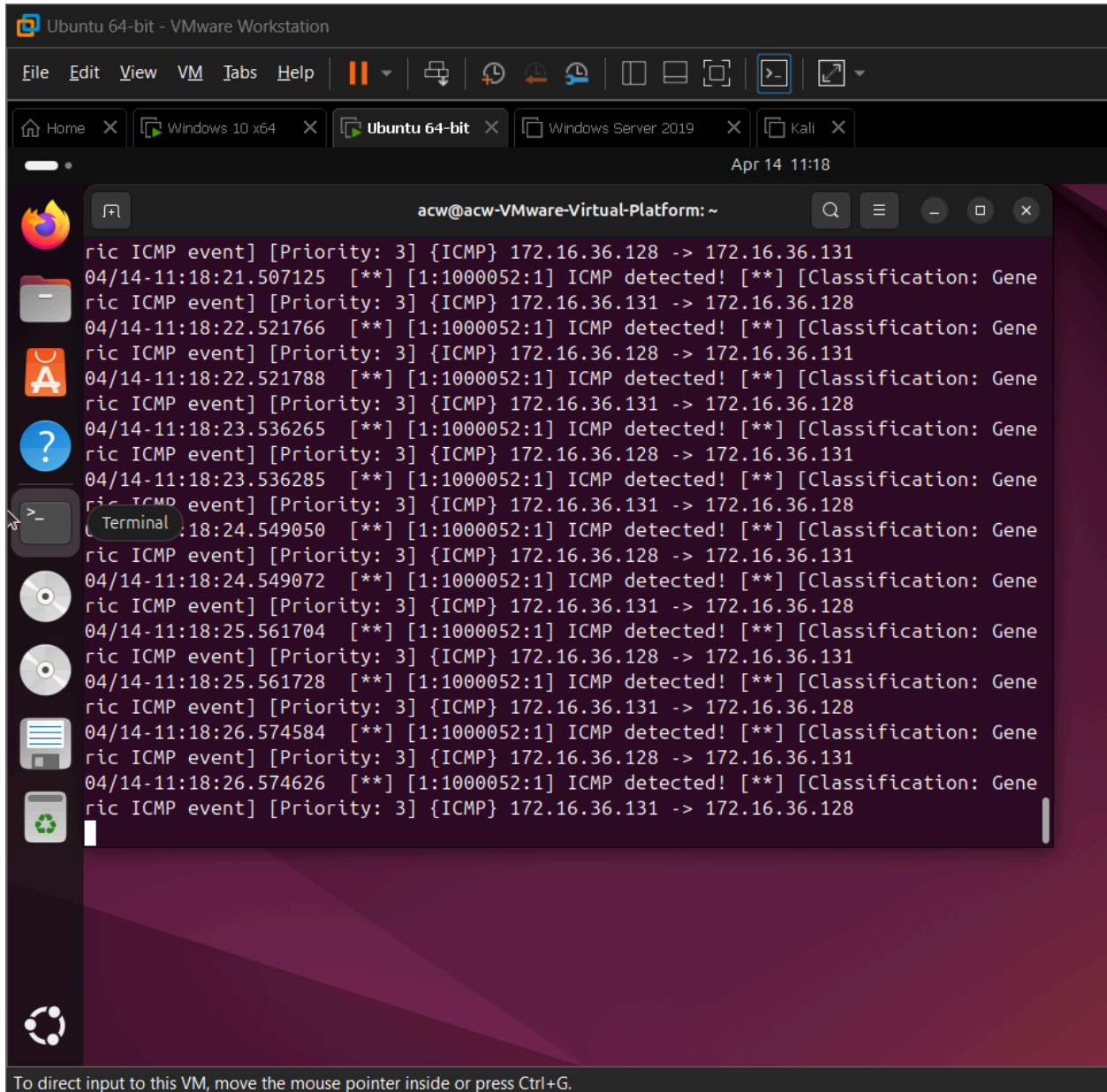


1c

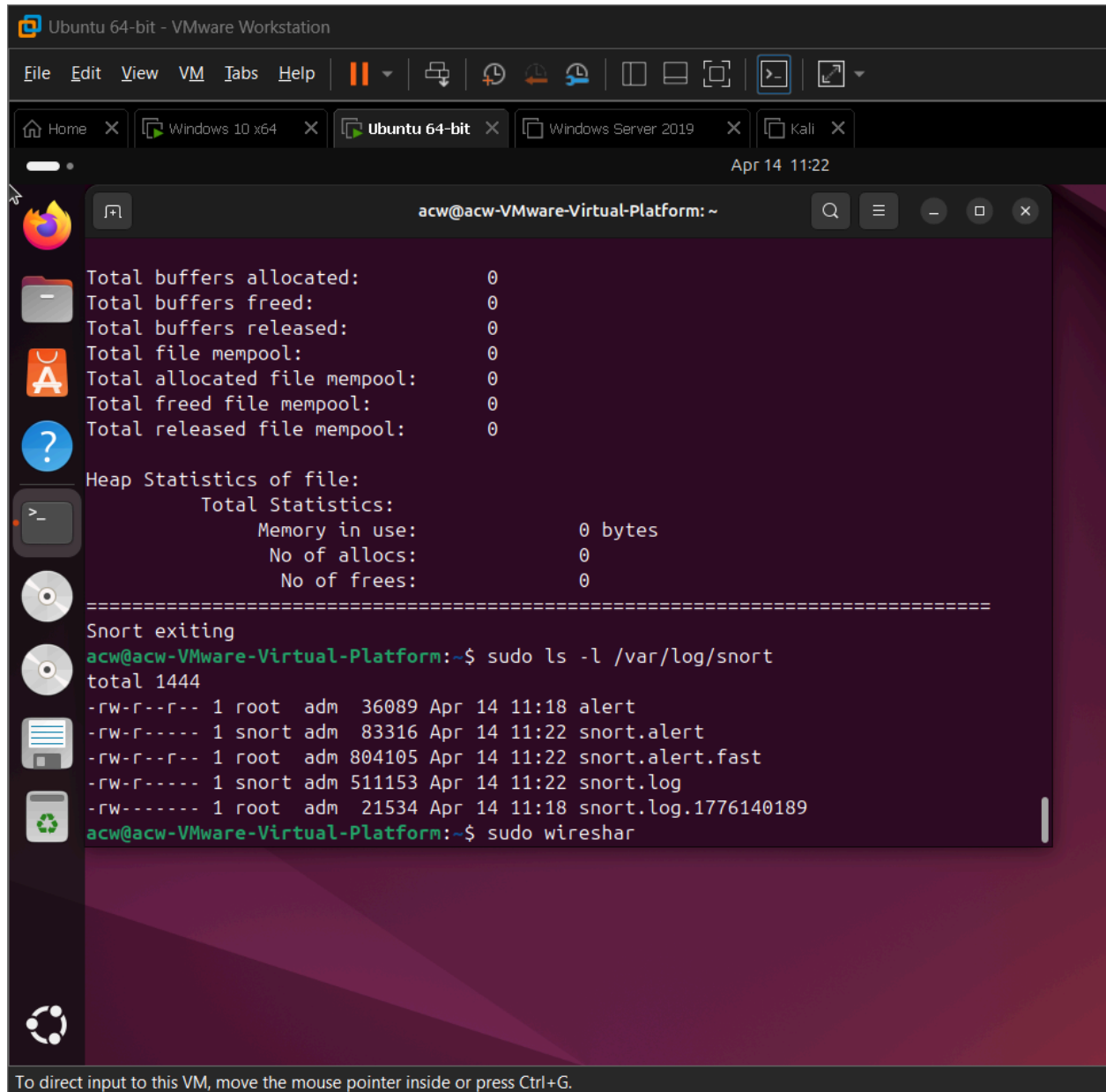


1d





le



1f

Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

Home Windows 10 x64 Ubuntu 64-bit Windows Server 2019 Kali

Apr 14 11:25

snort.log.1776140189

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
2	0.000032	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
3	1.012216	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
4	1.012237	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
5	2.031340	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
6	2.031369	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
7	3.314709	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
8	3.314731	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
9	4.325237	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
10	4.325267	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
11	5.335709	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request
12	5.335741	172.16.36.131	172.16.36.128	ICMP	74	Echo (ping) reply
13	6.345456	172.16.36.128	172.16.36.131	ICMP	74	Echo (ping) request

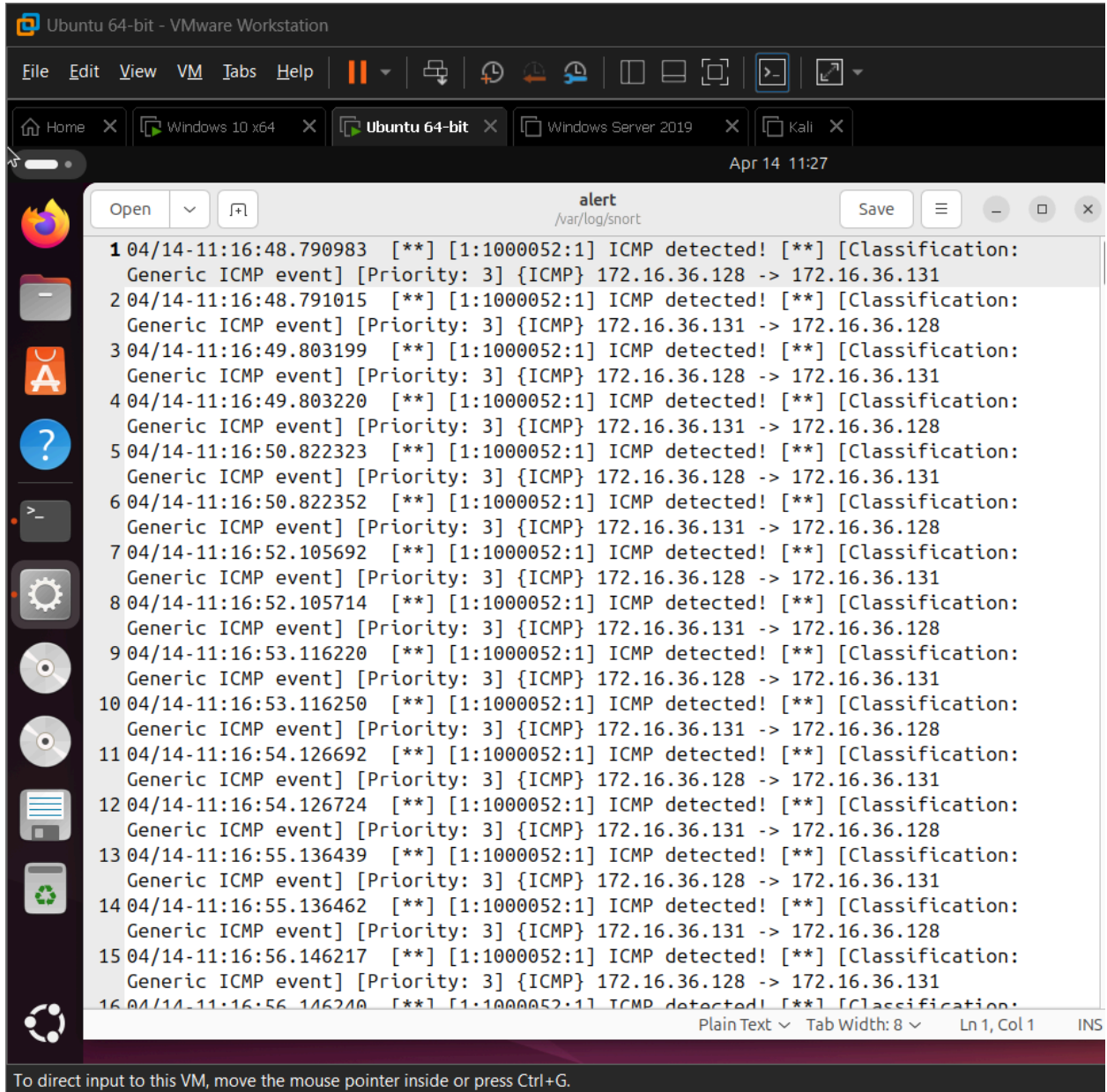
Frame 1: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface 0
Ethernet II, Src: VMware_a8:b4:29 (00:0c:29:a8:b4:29), Dst: VMware_a8:b4:29 (00:0c:29:a8:b4:29)
Internet Protocol Version 4, Src: 172.16.36.128, Dst: 172.16.36.131
Internet Control Message Protocol

0000 00 0c 29 87 02 7a 00 0c
0010 00 3c d5 79 00 00 80 01
0020 24 83 08 00 4c 5b 00 01
0030 67 68 69 6a 6b 6c 6d 6e
0040 77 61 62 63 64 65 66 67

snort.log.1776140189 Packets: 239 · Display

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

lg



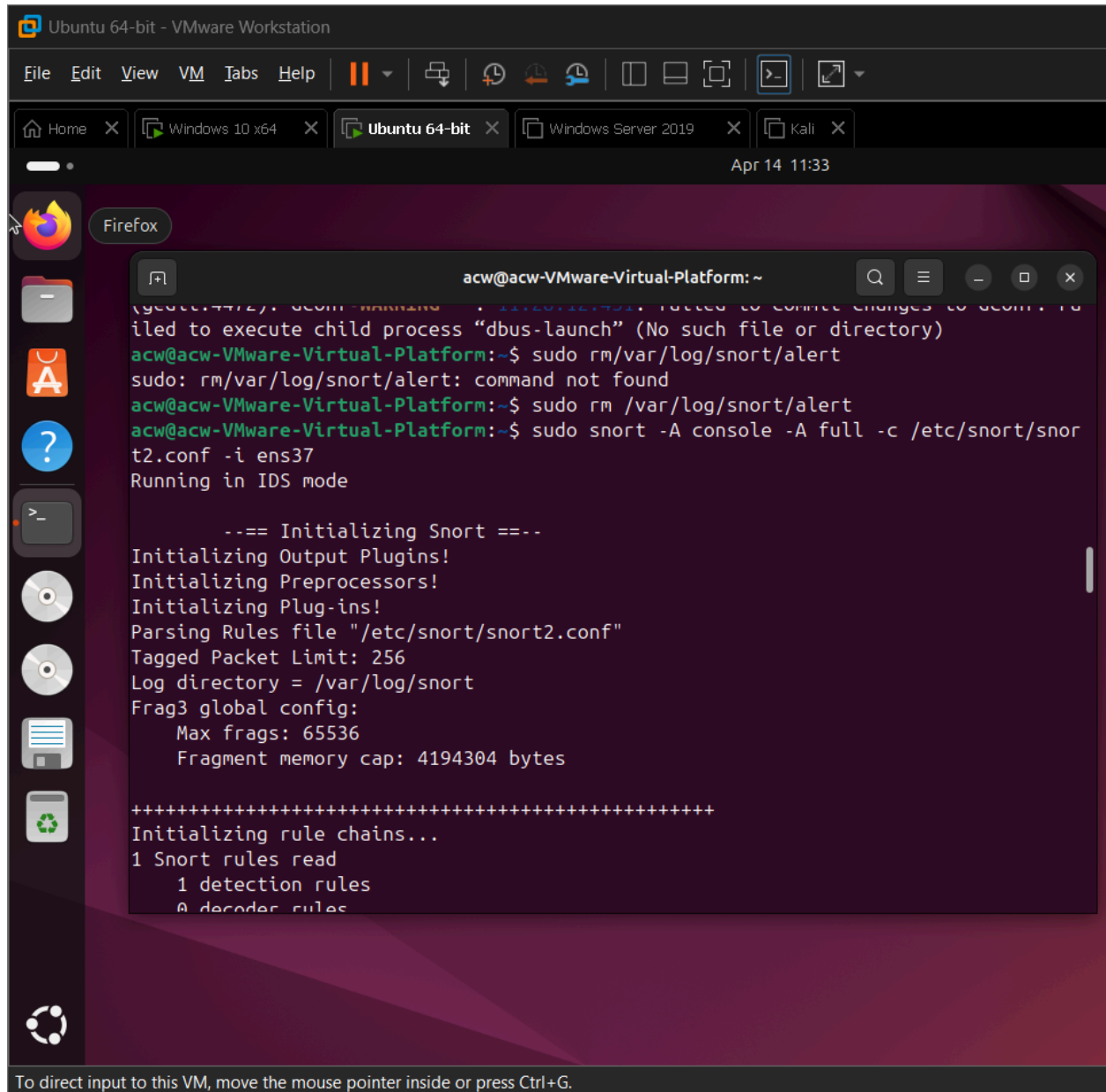
The screenshot shows a VMware Workstation interface with several virtual machines (VMs) open: Home, Windows 10 x64, Ubuntu 64-bit (selected), Windows Server 2019, and Kali. The Ubuntu 64-bit VM is running, and a terminal window titled 'alert' is open, displaying a log of ICMP detection events. The log entries are numbered 1 through 16, showing timestamps, IP addresses, and classification details. The terminal window has a menu bar with 'Open', 'Save', and window controls. The status bar at the bottom indicates 'Plain Text', 'Tab Width: 8', 'Ln 1, Col 1', and 'INS'.

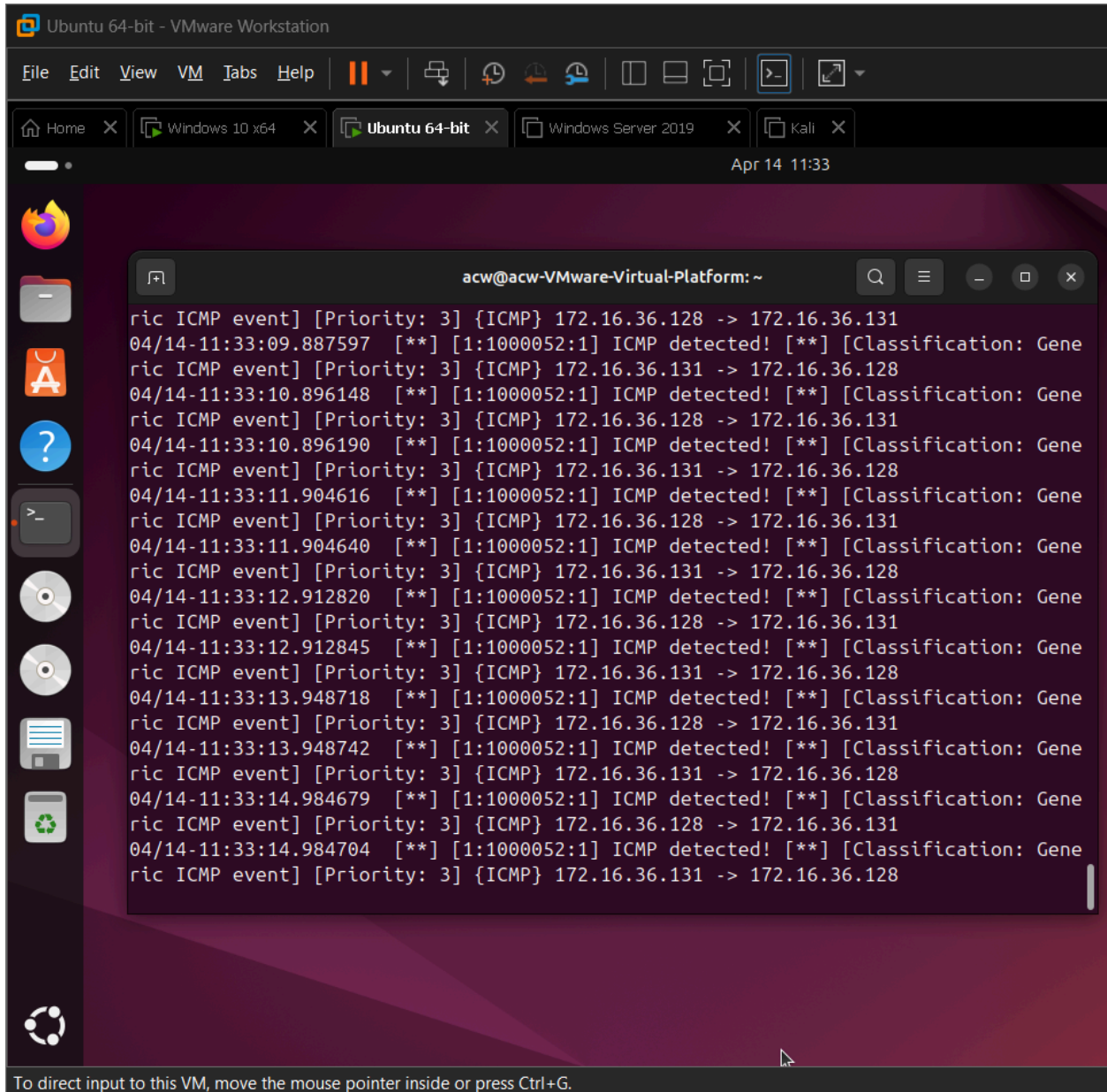
```
1 04/14-11:16:48.790983  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
2 04/14-11:16:48.791015  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
3 04/14-11:16:49.803199  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
4 04/14-11:16:49.803220  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
5 04/14-11:16:50.822323  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
6 04/14-11:16:50.822352  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
7 04/14-11:16:52.105692  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
8 04/14-11:16:52.105714  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
9 04/14-11:16:53.116220  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
10 04/14-11:16:53.116250 [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
11 04/14-11:16:54.126692  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
12 04/14-11:16:54.126724 [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
13 04/14-11:16:55.136439  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
14 04/14-11:16:55.136462  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
15 04/14-11:16:56.146217  [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
16 04/14-11:16:56.146240 [**] [1:1000052:1] ICMP detected! [**] [Classification:
```

Plain Text ▾ Tab Width: 8 ▾ Ln 1, Col 1 INS

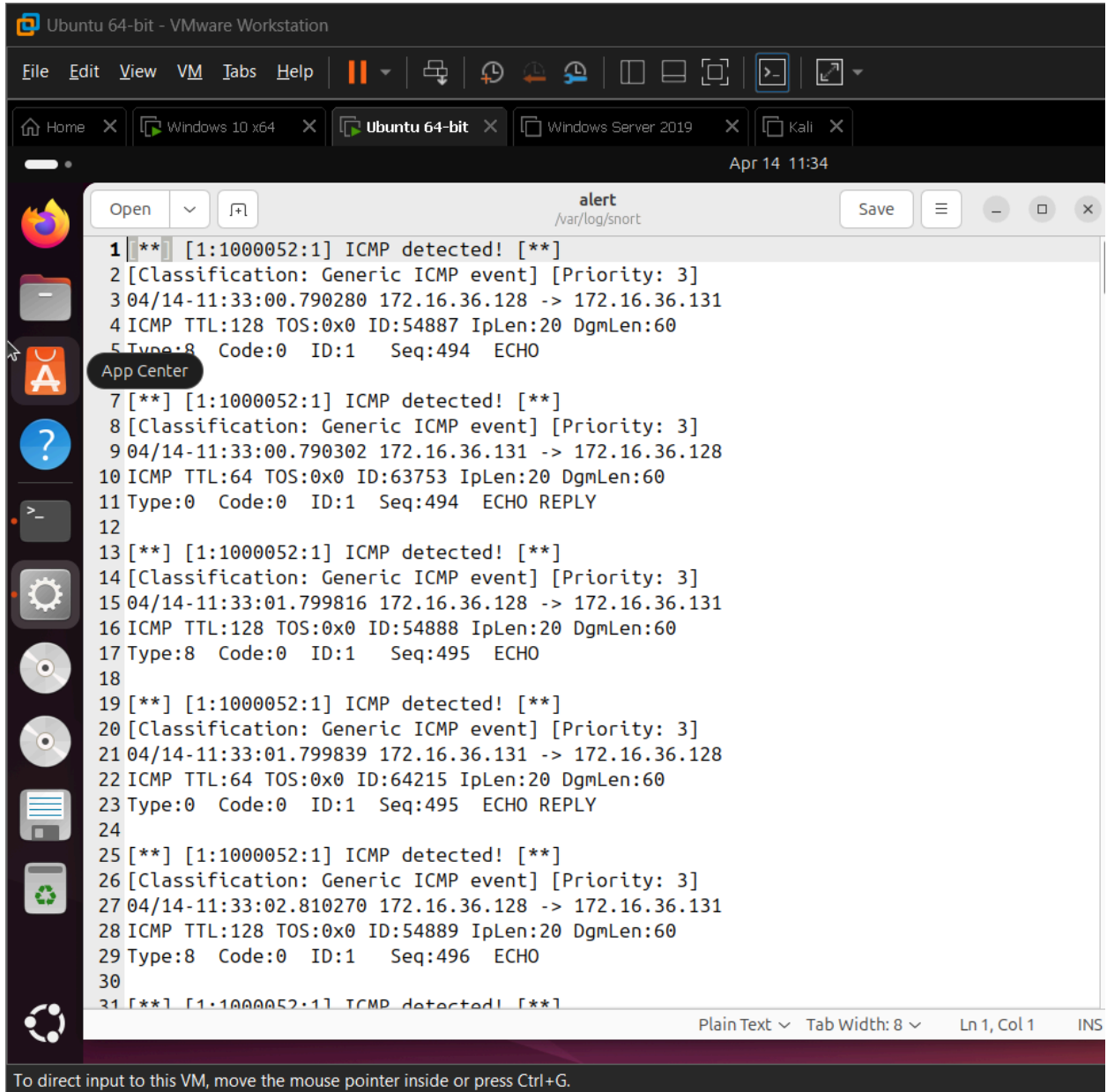
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

1h





li



Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

Home Windows 10 x64 Ubuntu 64-bit Windows Server 2019 Kali

Apr 14 11:34

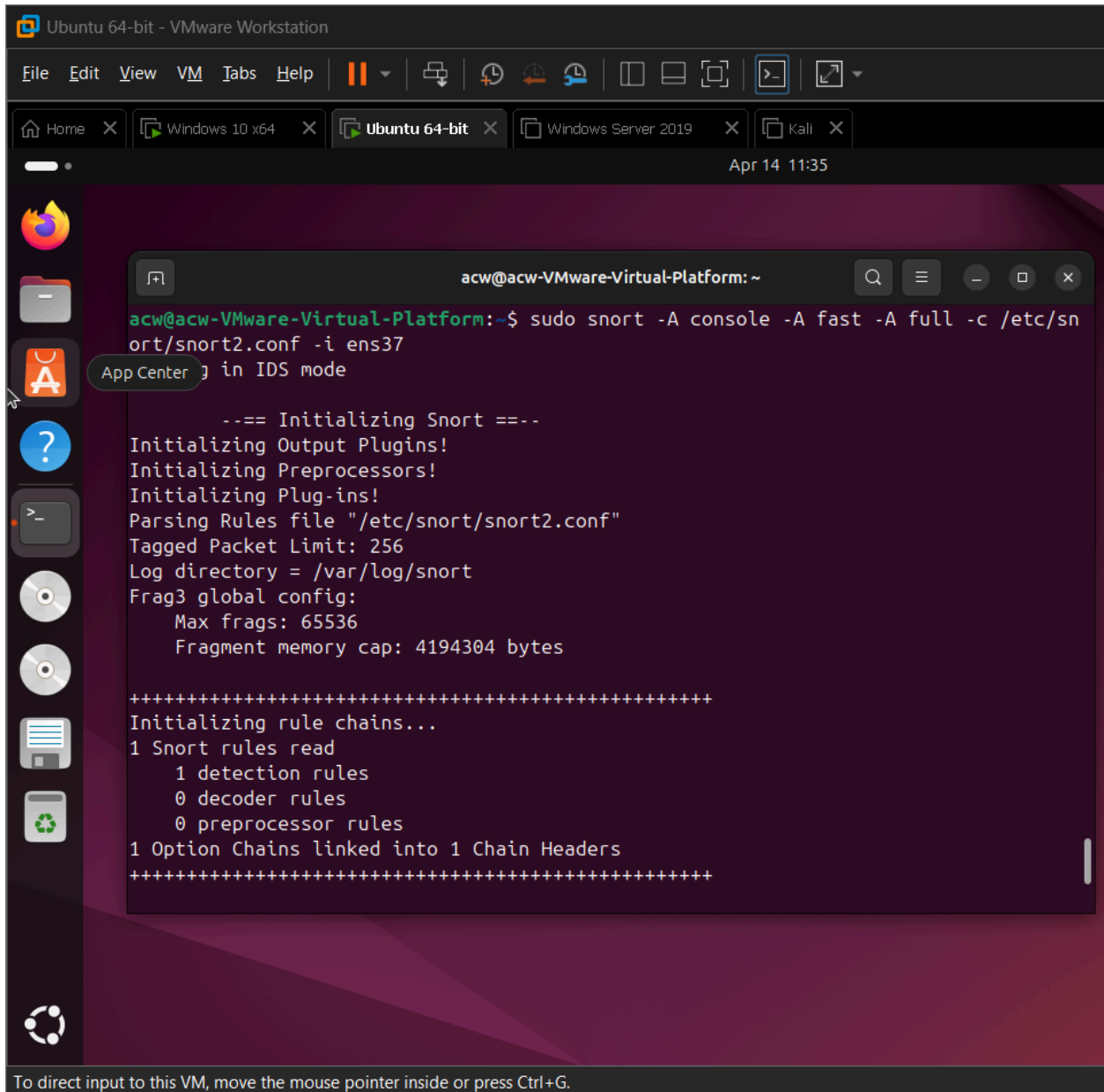
Open alert /var/log/snort Save

```
1 [**] [1:1000052:1] ICMP detected! [**]
2 [Classification: Generic ICMP event] [Priority: 3]
3 04/14-11:33:00.790280 172.16.36.128 -> 172.16.36.131
4 ICMP TTL:128 TOS:0x0 ID:54887 IpLen:20 DgmLen:60
5 Type:8 Code:0 ID:1 Seq:494 ECHO
6
7 [**] [1:1000052:1] ICMP detected! [**]
8 [Classification: Generic ICMP event] [Priority: 3]
9 04/14-11:33:00.790302 172.16.36.131 -> 172.16.36.128
10 ICMP TTL:64 TOS:0x0 ID:63753 IpLen:20 DgmLen:60
11 Type:0 Code:0 ID:1 Seq:494 ECHO REPLY
12
13 [**] [1:1000052:1] ICMP detected! [**]
14 [Classification: Generic ICMP event] [Priority: 3]
15 04/14-11:33:01.799816 172.16.36.128 -> 172.16.36.131
16 ICMP TTL:128 TOS:0x0 ID:54888 IpLen:20 DgmLen:60
17 Type:8 Code:0 ID:1 Seq:495 ECHO
18
19 [**] [1:1000052:1] ICMP detected! [**]
20 [Classification: Generic ICMP event] [Priority: 3]
21 04/14-11:33:01.799839 172.16.36.131 -> 172.16.36.128
22 ICMP TTL:64 TOS:0x0 ID:64215 IpLen:20 DgmLen:60
23 Type:0 Code:0 ID:1 Seq:495 ECHO REPLY
24
25 [**] [1:1000052:1] ICMP detected! [**]
26 [Classification: Generic ICMP event] [Priority: 3]
27 04/14-11:33:02.810270 172.16.36.128 -> 172.16.36.131
28 ICMP TTL:128 TOS:0x0 ID:54889 IpLen:20 DgmLen:60
29 Type:8 Code:0 ID:1 Seq:496 ECHO
30
31 [**] [1:1000052:1] ICMP detected! [**]
```

Plain Text Tab Width: 8 Ln 1, Col 1 INS

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

lj



Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

Home X Windows 10 x64 X Ubuntu 64-bit X Windows Server 2019 X Kali X

Apr 14 11:35

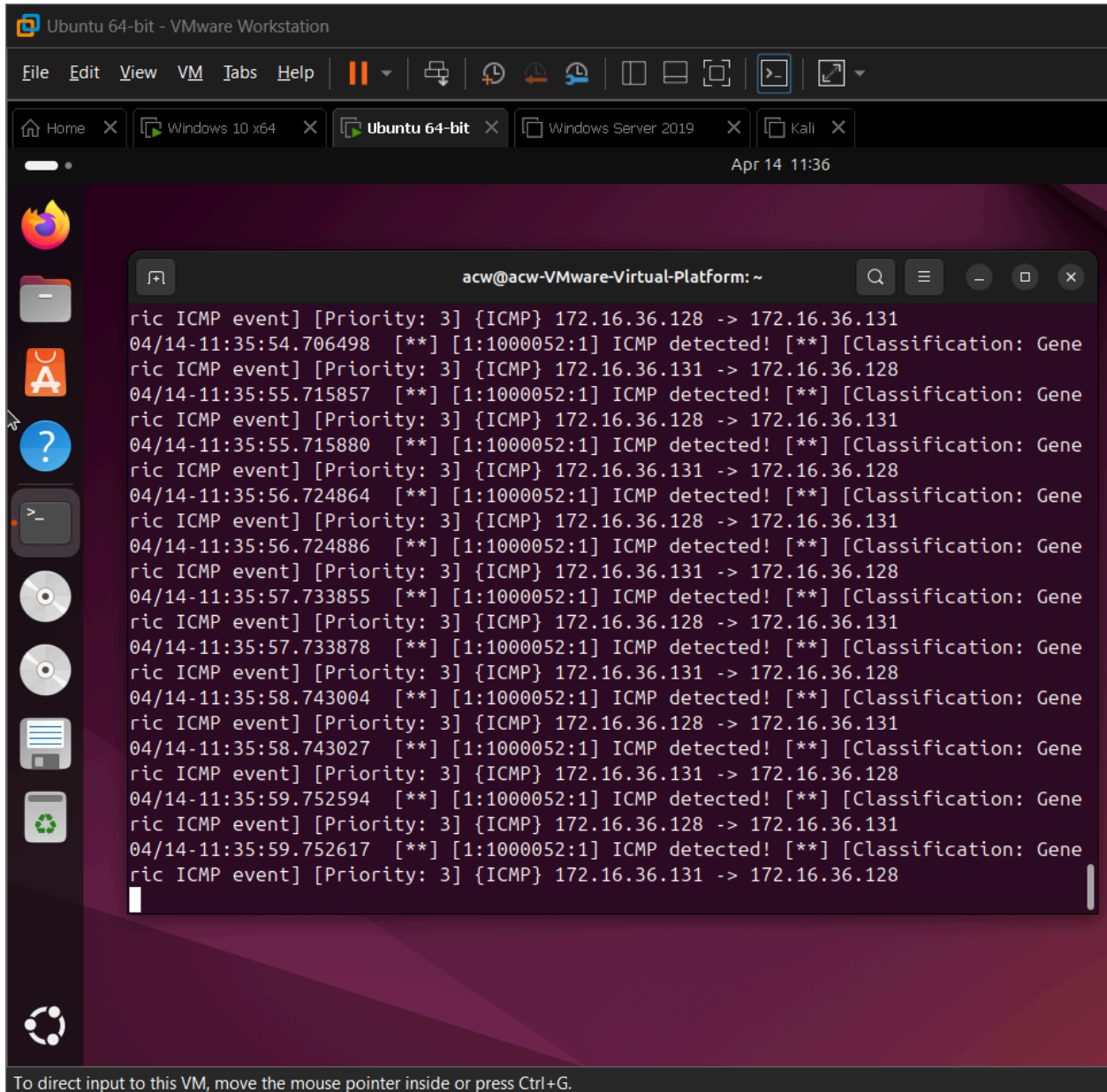
acw@acw-VMware-Virtual-Platform: ~

```
acw@acw-VMware-Virtual-Platform:~$ sudo snort -A console -A fast -A full -c /etc/snort/snort2.conf -i ens37
App Center in IDS mode

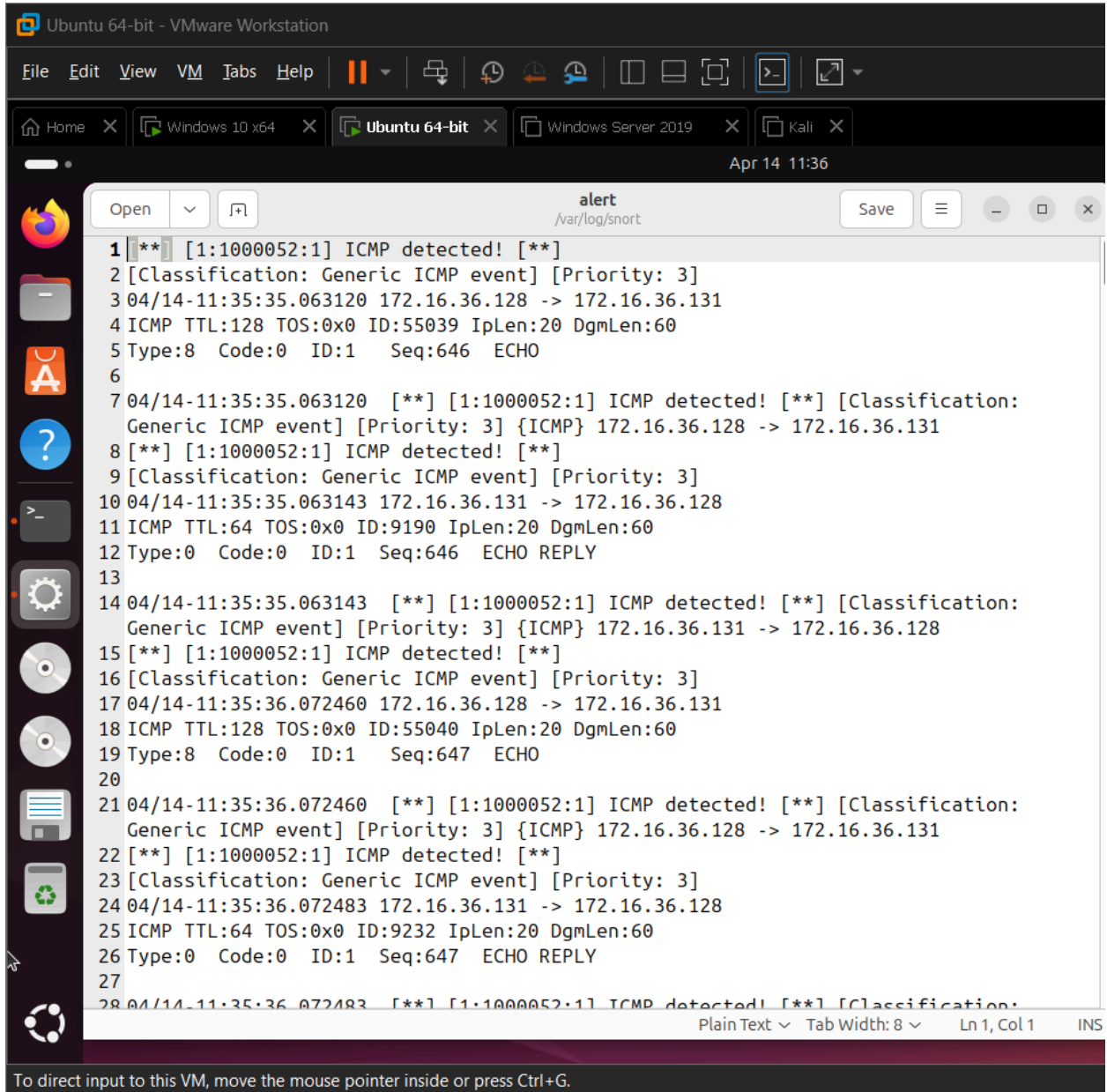
--== Initializing Snort ==--
Initializing Output Plugins!
Initializing Preprocessors!
Initializing Plug-ins!
Parsing Rules file "/etc/snort/snort2.conf"
Tagged Packet Limit: 256
Log directory = /var/log/snort
Frag3 global config:
    Max frags: 65536
    Fragment memory cap: 4194304 bytes

+++++
Initializing rule chains...
1 Snort rules read
    1 detection rules
    0 decoder rules
    0 preprocessor rules
1 Option Chains linked into 1 Chain Headers
+++++
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



1k



Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

Home X Windows 10 x64 X Ubuntu 64-bit X Windows Server 2019 X Kali X

Apr 14 11:36

Open alert Save

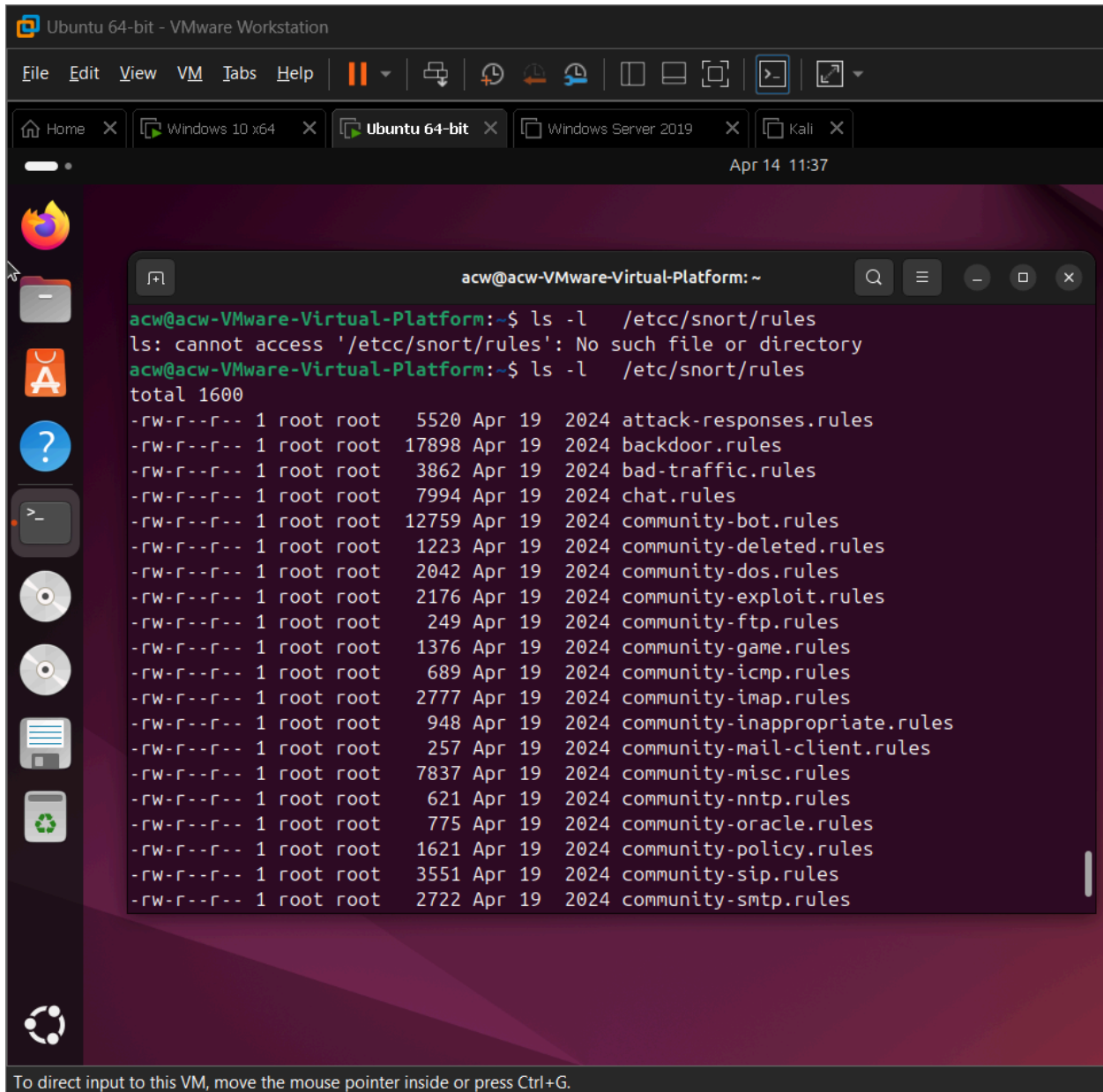
/var/log/snort

```
1 [**] [1:1000052:1] ICMP detected! [**]
2 [Classification: Generic ICMP event] [Priority: 3]
3 04/14-11:35:35.063120 172.16.36.128 -> 172.16.36.131
4 ICMP TTL:128 TOS:0x0 ID:55039 IpLen:20 DgmLen:60
5 Type:8 Code:0 ID:1 Seq:646 ECHO
6
7 04/14-11:35:35.063120 [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
8 [**] [1:1000052:1] ICMP detected! [**]
9 [Classification: Generic ICMP event] [Priority: 3]
10 04/14-11:35:35.063143 172.16.36.131 -> 172.16.36.128
11 ICMP TTL:64 TOS:0x0 ID:9190 IpLen:20 DgmLen:60
12 Type:0 Code:0 ID:1 Seq:646 ECHO REPLY
13
14 04/14-11:35:35.063143 [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
15 [**] [1:1000052:1] ICMP detected! [**]
16 [Classification: Generic ICMP event] [Priority: 3]
17 04/14-11:35:36.072460 172.16.36.128 -> 172.16.36.131
18 ICMP TTL:128 TOS:0x0 ID:55040 IpLen:20 DgmLen:60
19 Type:8 Code:0 ID:1 Seq:647 ECHO
20
21 04/14-11:35:36.072460 [**] [1:1000052:1] ICMP detected! [**] [Classification:
  Generic ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
22 [**] [1:1000052:1] ICMP detected! [**]
23 [Classification: Generic ICMP event] [Priority: 3]
24 04/14-11:35:36.072483 172.16.36.131 -> 172.16.36.128
25 ICMP TTL:64 TOS:0x0 ID:9232 IpLen:20 DgmLen:60
26 Type:0 Code:0 ID:1 Seq:647 ECHO REPLY
27
28 04/14-11:35:36.072483 [**] [1:1000052:1] ICMP detected! [**] [Classification:
```

Plain Text Tab Width: 8 Ln 1, Col 1 INS

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

2a



Ubuntu 64-bit - VMware Workstation

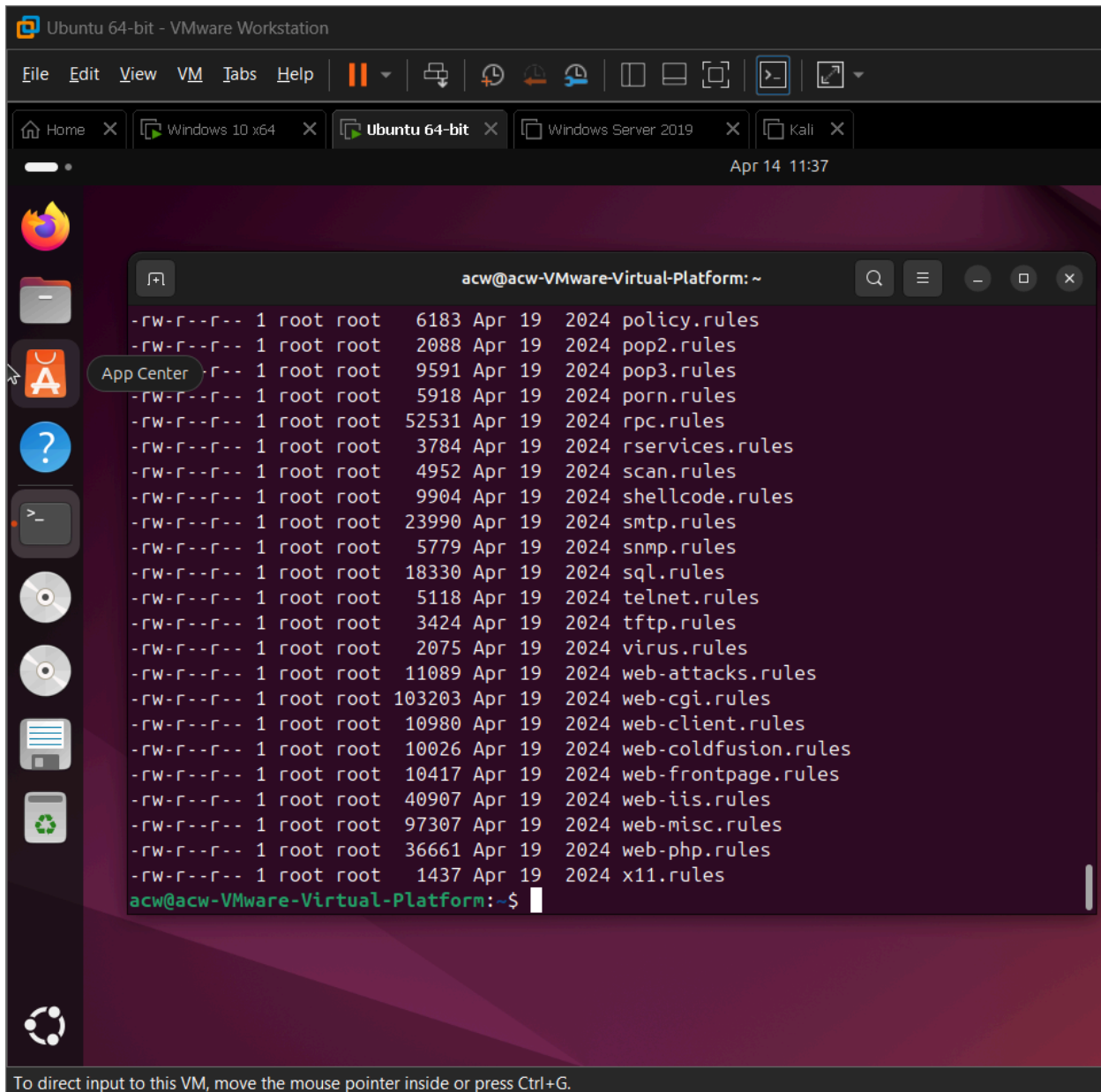
File Edit View VM Tabs Help

Home Windows 10 x64 Ubuntu 64-bit Windows Server 2019 Kali

Apr 14 11:37

```
acw@acw-VMware-Virtual-Platform: ~  
acw@acw-VMware-Virtual-Platform:~$ ls -l /etcc/snort/rules  
ls: cannot access '/etcc/snort/rules': No such file or directory  
acw@acw-VMware-Virtual-Platform:~$ ls -l /etc/snort/rules  
total 1600  
-rw-r--r-- 1 root root 5520 Apr 19 2024 attack-responses.rules  
-rw-r--r-- 1 root root 17898 Apr 19 2024 backdoor.rules  
-rw-r--r-- 1 root root 3862 Apr 19 2024 bad-traffic.rules  
-rw-r--r-- 1 root root 7994 Apr 19 2024 chat.rules  
-rw-r--r-- 1 root root 12759 Apr 19 2024 community-bot.rules  
-rw-r--r-- 1 root root 1223 Apr 19 2024 community-deleted.rules  
-rw-r--r-- 1 root root 2042 Apr 19 2024 community-dos.rules  
-rw-r--r-- 1 root root 2176 Apr 19 2024 community-exploit.rules  
-rw-r--r-- 1 root root 249 Apr 19 2024 community-ftp.rules  
-rw-r--r-- 1 root root 1376 Apr 19 2024 community-game.rules  
-rw-r--r-- 1 root root 689 Apr 19 2024 community-icmp.rules  
-rw-r--r-- 1 root root 2777 Apr 19 2024 community-imap.rules  
-rw-r--r-- 1 root root 948 Apr 19 2024 community-inappropriate.rules  
-rw-r--r-- 1 root root 257 Apr 19 2024 community-mail-client.rules  
-rw-r--r-- 1 root root 7837 Apr 19 2024 community-misc.rules  
-rw-r--r-- 1 root root 621 Apr 19 2024 community-nntp.rules  
-rw-r--r-- 1 root root 775 Apr 19 2024 community-oracle.rules  
-rw-r--r-- 1 root root 1621 Apr 19 2024 community-policy.rules  
-rw-r--r-- 1 root root 3551 Apr 19 2024 community-sip.rules  
-rw-r--r-- 1 root root 2722 Apr 19 2024 community-smtp.rules
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



2c: Which five categories are the most interesting?

Personally, I would choose these five:

exploit.rules: Detects known "hacks" against software vulnerabilities.

malware-cnc.rules: Identifies computers talking to "Command and Control" servers (Botnets).

dos.rules: Detects Denial of Service attacks meant to crash a system.

sql-injection.rules: Detects attempts to steal data from databases.

spyware-put.rules: Detects when your computer is trying to upload your private data to a thief.

2d: Which five individual rules are the most interesting?

Personally, I would choose:

ICMP PING NMAP

ICMP Large

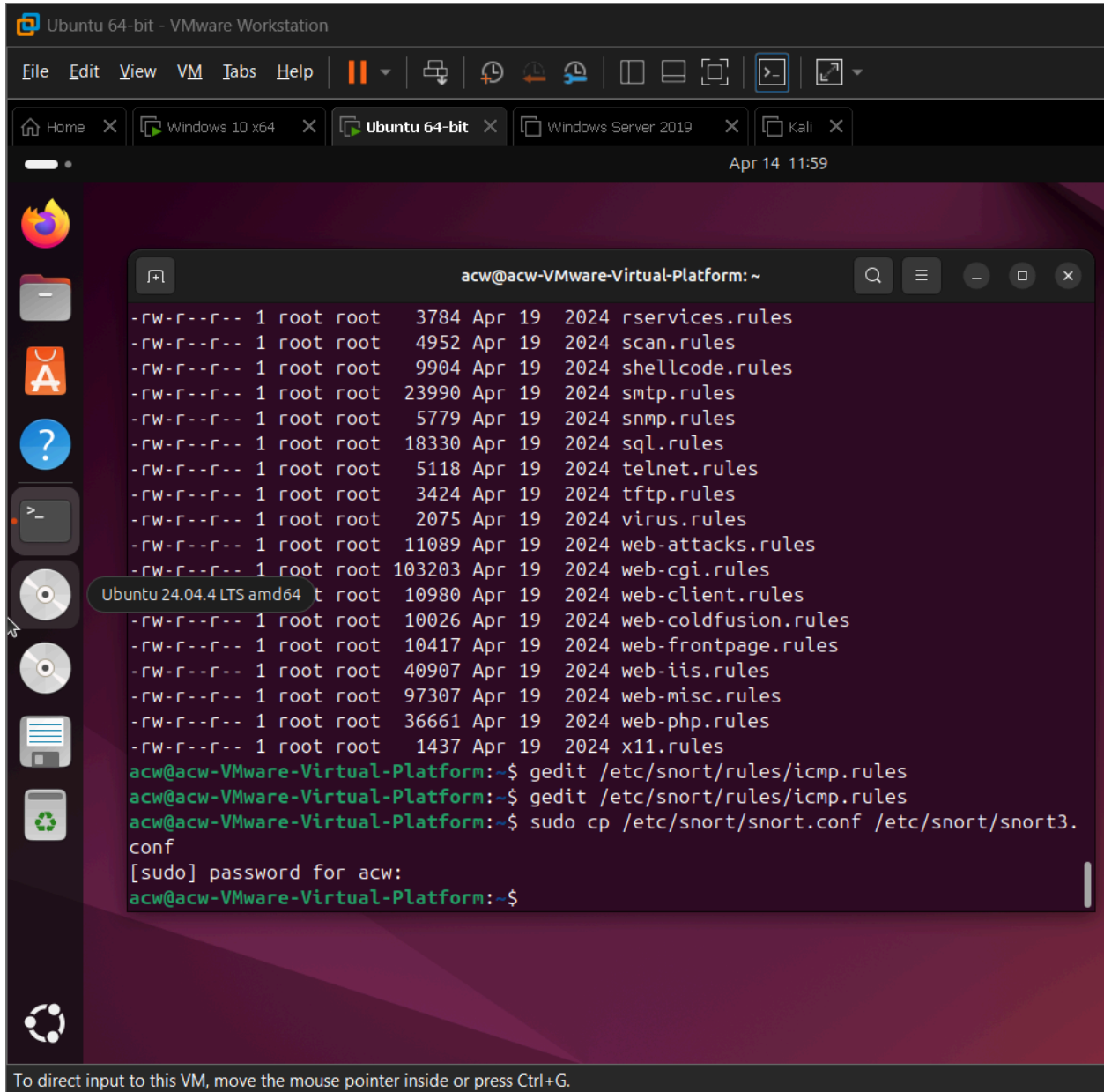
ICMP Nemesis v1.1 Echo

ICMP Redirect Host

ICMP icmpenum v1.1.1

3a-3e

3a



Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

Home Windows 10 x64 Ubuntu 64-bit Windows Server 2019 Kali

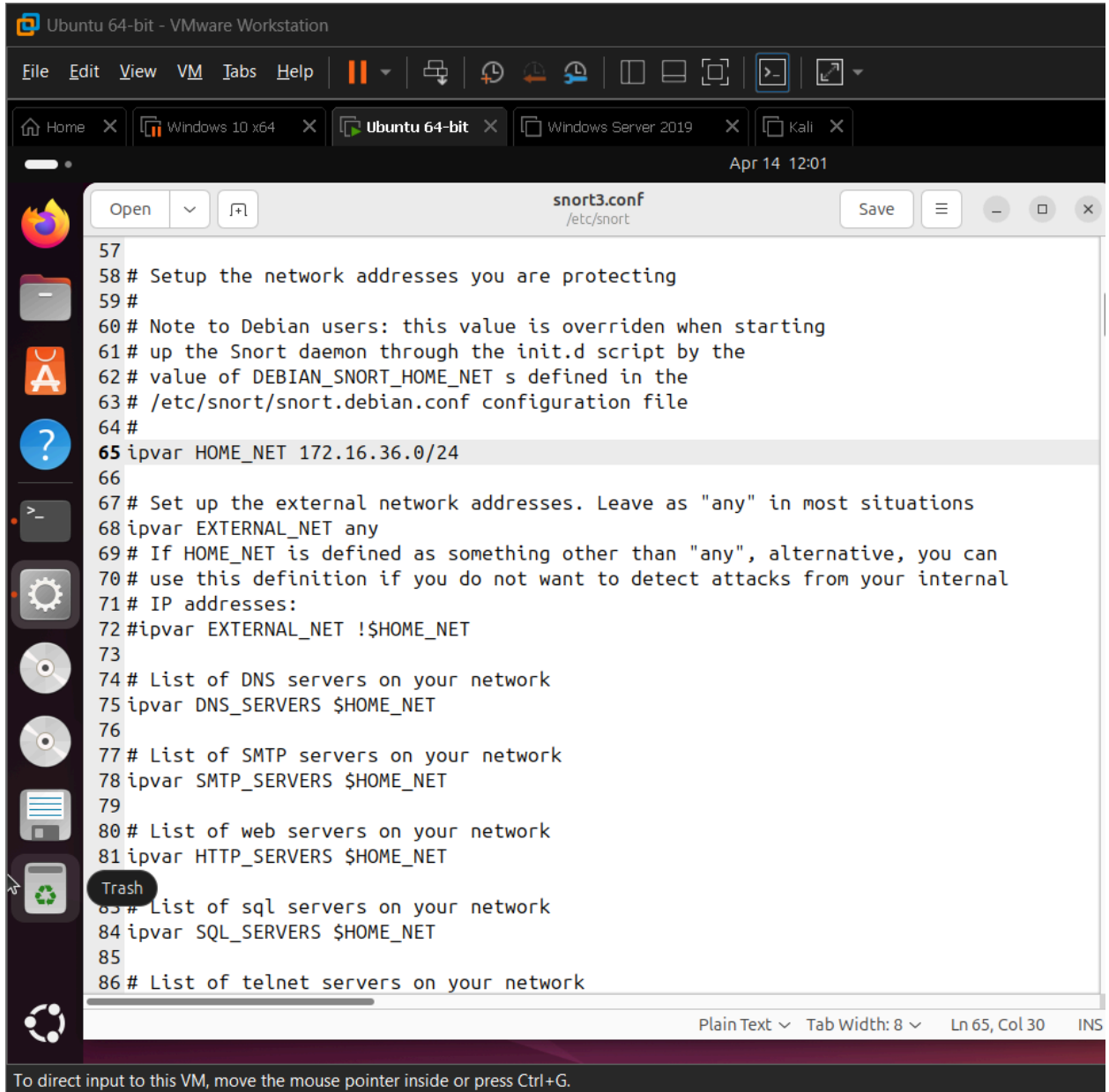
Apr 14 11:59

acw@acw-VMware-Virtual-Platform: ~

```
-rw-r--r-- 1 root root 3784 Apr 19 2024 rservices.rules
-rw-r--r-- 1 root root 4952 Apr 19 2024 scan.rules
-rw-r--r-- 1 root root 9904 Apr 19 2024 shellcode.rules
-rw-r--r-- 1 root root 23990 Apr 19 2024 smtp.rules
-rw-r--r-- 1 root root 5779 Apr 19 2024 snmp.rules
-rw-r--r-- 1 root root 18330 Apr 19 2024 sql.rules
-rw-r--r-- 1 root root 5118 Apr 19 2024 telnet.rules
-rw-r--r-- 1 root root 3424 Apr 19 2024 tftp.rules
-rw-r--r-- 1 root root 2075 Apr 19 2024 virus.rules
-rw-r--r-- 1 root root 11089 Apr 19 2024 web-attacks.rules
-rw-r--r-- 1 root root 103203 Apr 19 2024 web-cgi.rules
Ubuntu 24.04.4 LTS amd64 t root 10980 Apr 19 2024 web-client.rules
-rw-r--r-- 1 root root 10026 Apr 19 2024 web-coldfusion.rules
-rw-r--r-- 1 root root 10417 Apr 19 2024 web-frontpage.rules
-rw-r--r-- 1 root root 40907 Apr 19 2024 web-iis.rules
-rw-r--r-- 1 root root 97307 Apr 19 2024 web-misc.rules
-rw-r--r-- 1 root root 36661 Apr 19 2024 web-php.rules
-rw-r--r-- 1 root root 1437 Apr 19 2024 x11.rules
acw@acw-VMware-Virtual-Platform:~$ gedit /etc/snort/rules/icmp.rules
acw@acw-VMware-Virtual-Platform:~$ gedit /etc/snort/rules/icmp.rules
acw@acw-VMware-Virtual-Platform:~$ sudo cp /etc/snort/snort.conf /etc/snort/snort3.conf
[sudo] password for acw:
acw@acw-VMware-Virtual-Platform:~$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

3b



Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

Home Windows 10 x64 Ubuntu 64-bit Windows Server 2019 Kali

Apr 14 12:01

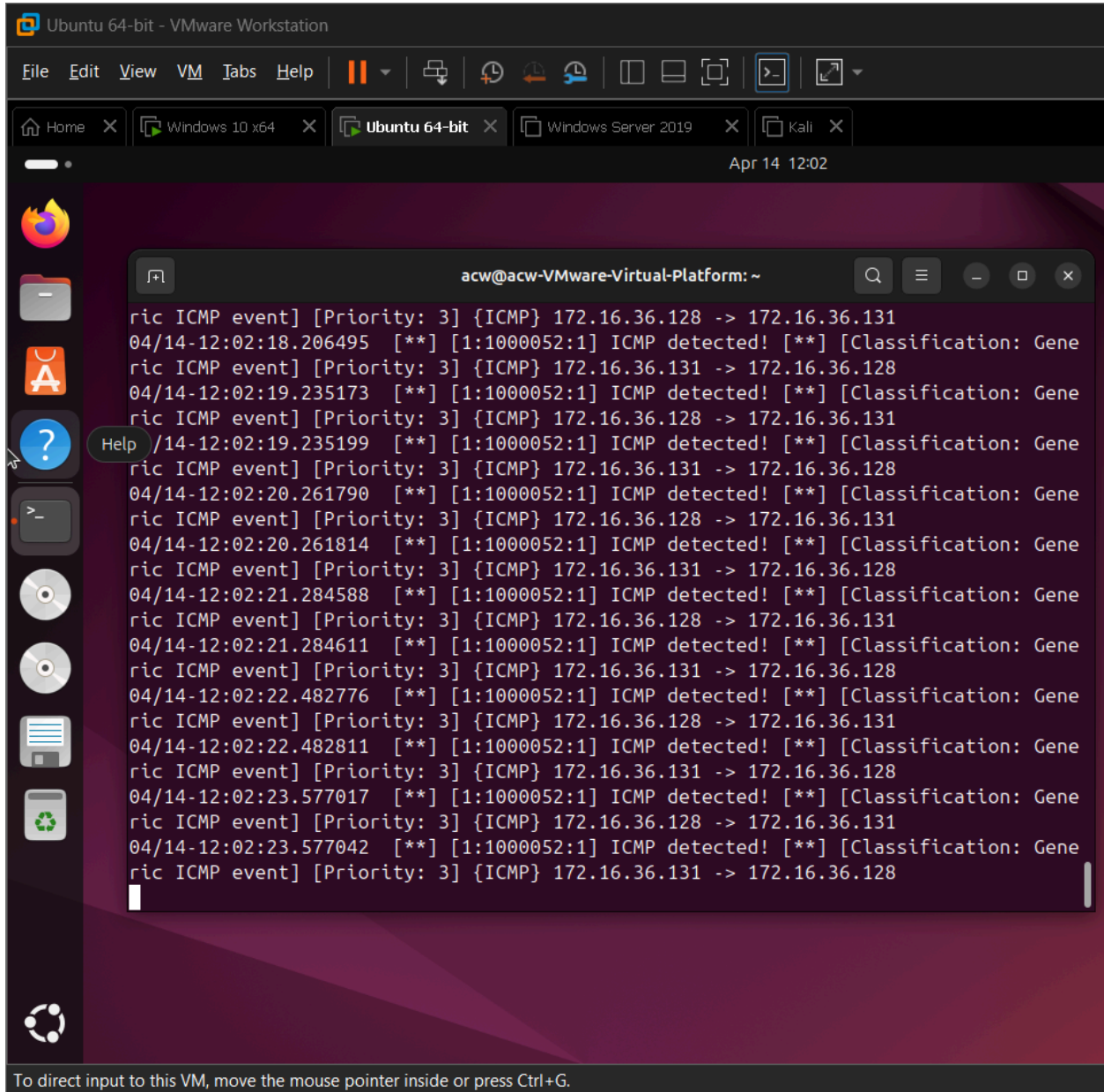
Open snort3.conf /etc/snort Save

```
57
58 # Setup the network addresses you are protecting
59 #
60 # Note to Debian users: this value is overridden when starting
61 # up the Snort daemon through the init.d script by the
62 # value of DEBIAN_SNORT_HOME_NET s defined in the
63 # /etc/snort/snort.debian.conf configuration file
64 #
65 ipvar HOME_NET 172.16.36.0/24
66
67 # Set up the external network addresses. Leave as "any" in most situations
68 ipvar EXTERNAL_NET any
69 # If HOME_NET is defined as something other than "any", alternative, you can
70 # use this definition if you do not want to detect attacks from your internal
71 # IP addresses:
72 #ipvar EXTERNAL_NET !$HOME_NET
73
74 # List of DNS servers on your network
75 ipvar DNS_SERVERS $HOME_NET
76
77 # List of SMTP servers on your network
78 ipvar SMTP_SERVERS $HOME_NET
79
80 # List of web servers on your network
81 ipvar HTTP_SERVERS $HOME_NET
82
83 # List of sql servers on your network
84 ipvar SQL_SERVERS $HOME_NET
85
86 # List of telnet servers on your network
```

Plain Text Tab Width: 8 Ln 65, Col 30 INS

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

3c



Ubuntu 64-bit - VMware Workstation

File Edit View VM Tabs Help

Home Windows 10 x64 Ubuntu 64-bit Windows Server 2019 Kali

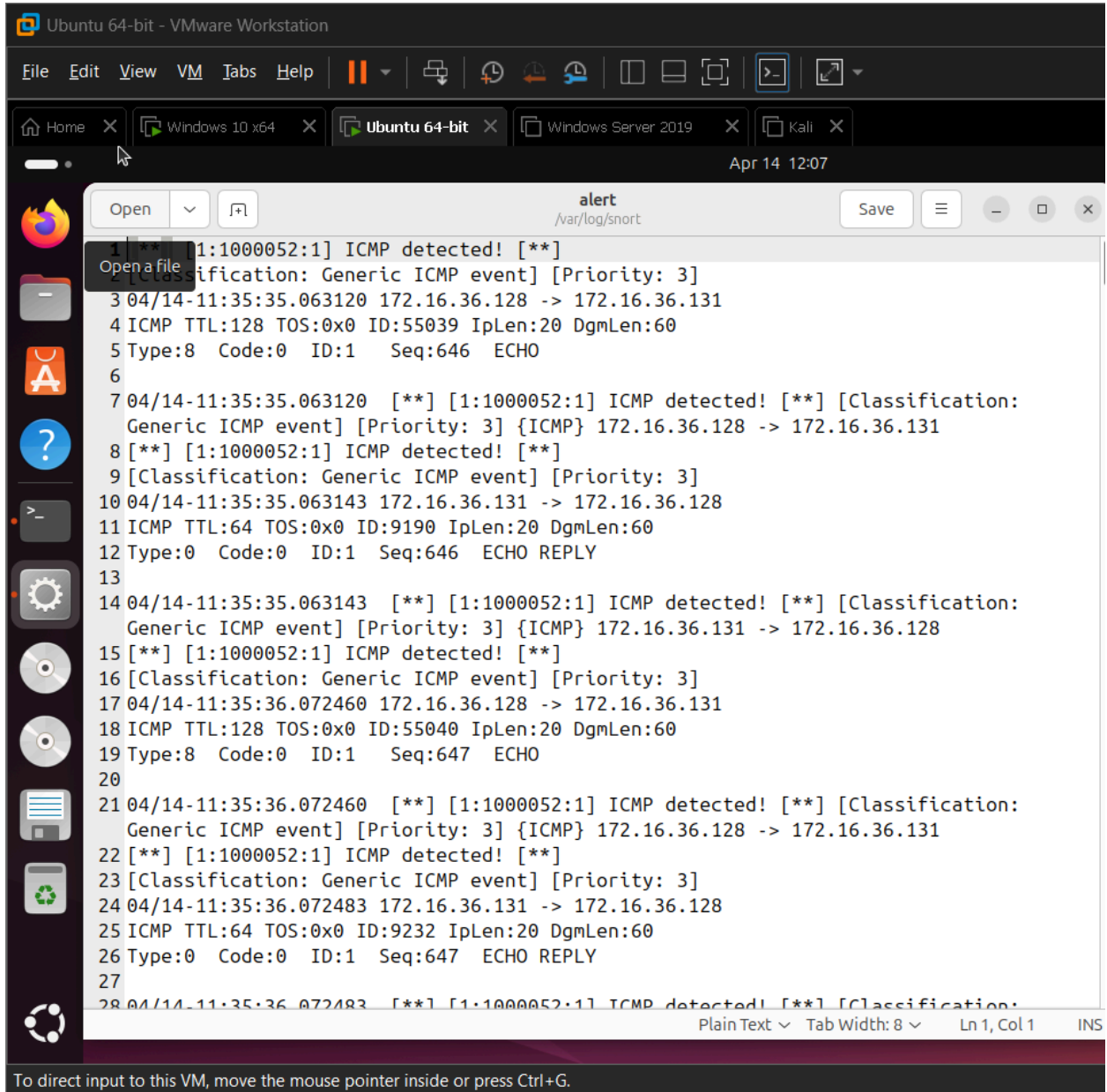
Apr 14 12:02

acw@acw-VMware-Virtual-Platform: ~

```
ric ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
04/14-12:02:18.206495  [**] [1:1000052:1] ICMP detected! [**] [Classification: Gene
ric ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
04/14-12:02:19.235173  [**] [1:1000052:1] ICMP detected! [**] [Classification: Gene
ric ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
04/14-12:02:19.235199  [**] [1:1000052:1] ICMP detected! [**] [Classification: Gene
ric ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
04/14-12:02:20.261790  [**] [1:1000052:1] ICMP detected! [**] [Classification: Gene
ric ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
04/14-12:02:20.261814  [**] [1:1000052:1] ICMP detected! [**] [Classification: Gene
ric ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
04/14-12:02:21.284588  [**] [1:1000052:1] ICMP detected! [**] [Classification: Gene
ric ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
04/14-12:02:21.284611  [**] [1:1000052:1] ICMP detected! [**] [Classification: Gene
ric ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
04/14-12:02:22.482776  [**] [1:1000052:1] ICMP detected! [**] [Classification: Gene
ric ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
04/14-12:02:22.482811  [**] [1:1000052:1] ICMP detected! [**] [Classification: Gene
ric ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
04/14-12:02:23.577017  [**] [1:1000052:1] ICMP detected! [**] [Classification: Gene
ric ICMP event] [Priority: 3] {ICMP} 172.16.36.128 -> 172.16.36.131
04/14-12:02:23.577042  [**] [1:1000052:1] ICMP detected! [**] [Classification: Gene
ric ICMP event] [Priority: 3] {ICMP} 172.16.36.131 -> 172.16.36.128
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

3d



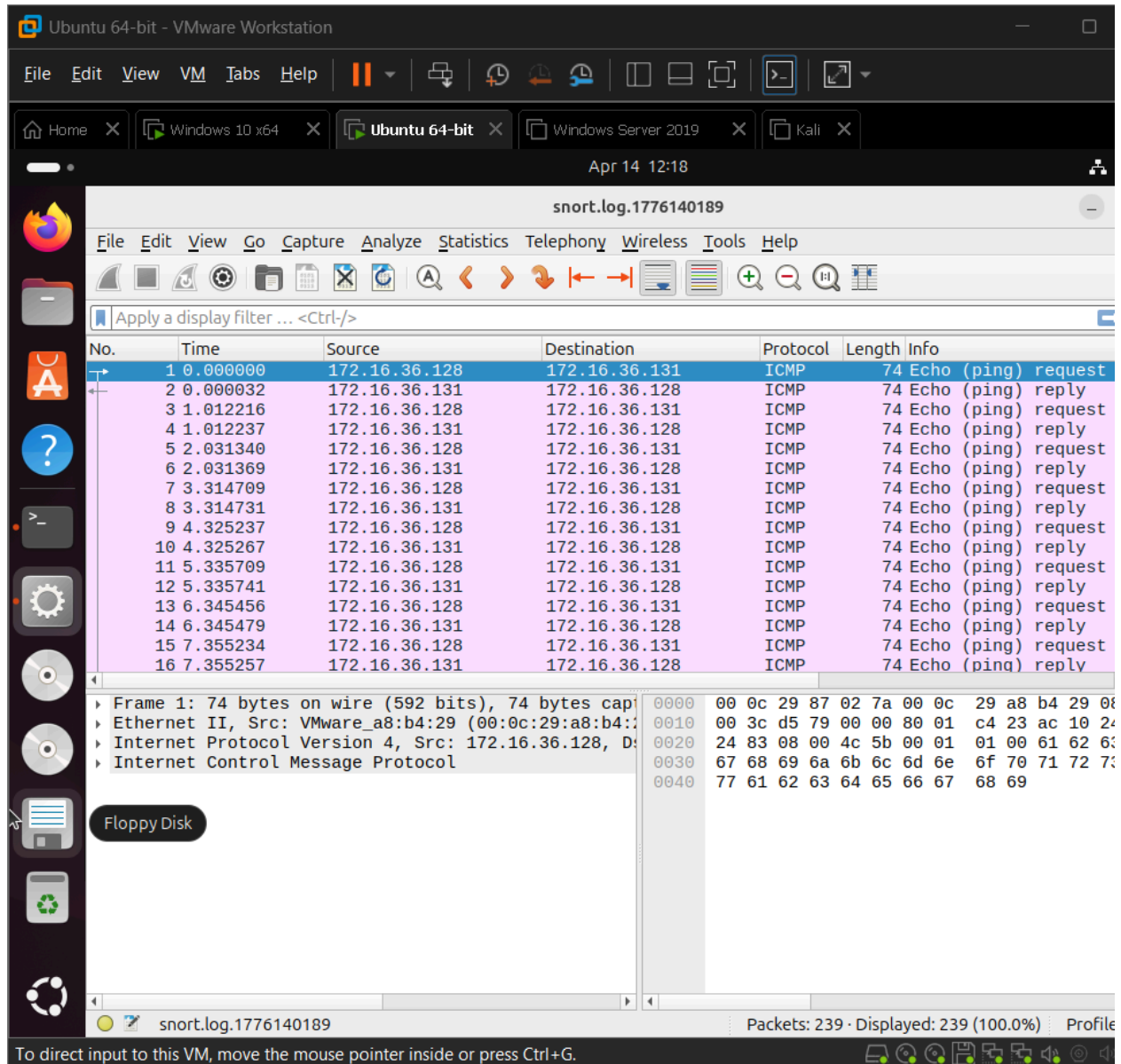
The screenshot shows a VMware Workstation interface with several virtual machines (VMs) open: Home, Windows 10 x64, Ubuntu 64-bit (selected), Windows Server 2019, and Kali. The Ubuntu 64-bit VM is running, and a terminal window titled 'alert' is open, displaying network traffic logs from the file /var/log/snort. The logs show ICMP events, including 'ICMP detected!' and 'ICMP TTL:128 TOS:0x0 ID:55039 IpLen:20 DgmLen:60'. The terminal window has a 'Save' button and a menu icon. The bottom status bar indicates 'Plain Text', 'Tab Width: 8', 'Ln 1, Col 1', and 'INS'.

```
1 1:1000052:1] ICMP detected! [**]
2 [Classification: Generic ICMP event] [Priority: 3]
3 04/14-11:35:35.063120 172.16.36.128 -> 172.16.36.131
4 ICMP TTL:128 TOS:0x0 ID:55039 IpLen:20 DgmLen:60
5 Type:8 Code:0 ID:1 Seq:646 ECHO
6
7 04/14-11:35:35.063120 [**] [1:1000052:1] ICMP detected! [**] [Classification:
8 [**] [1:1000052:1] ICMP detected! [**]
9 [Classification: Generic ICMP event] [Priority: 3]
10 04/14-11:35:35.063143 172.16.36.131 -> 172.16.36.128
11 ICMP TTL:64 TOS:0x0 ID:9190 IpLen:20 DgmLen:60
12 Type:0 Code:0 ID:1 Seq:646 ECHO REPLY
13
14 04/14-11:35:35.063143 [**] [1:1000052:1] ICMP detected! [**] [Classification:
15 [**] [1:1000052:1] ICMP detected! [**]
16 [Classification: Generic ICMP event] [Priority: 3]
17 04/14-11:35:36.072460 172.16.36.128 -> 172.16.36.131
18 ICMP TTL:128 TOS:0x0 ID:55040 IpLen:20 DgmLen:60
19 Type:8 Code:0 ID:1 Seq:647 ECHO
20
21 04/14-11:35:36.072460 [**] [1:1000052:1] ICMP detected! [**] [Classification:
22 [**] [1:1000052:1] ICMP detected! [**]
23 [Classification: Generic ICMP event] [Priority: 3]
24 04/14-11:35:36.072483 172.16.36.131 -> 172.16.36.128
25 ICMP TTL:64 TOS:0x0 ID:9232 IpLen:20 DgmLen:60
26 Type:0 Code:0 ID:1 Seq:647 ECHO REPLY
27
28 04/14-11:35:36.072483 [**] [1:1000052:1] ICMP detected! [**] [Classification:
```

Plain Text ▾ Tab Width: 8 ▾ Ln 1, Col 1 INS

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

3e



Lab Analysis

1. What are the three modes of Snort?

The three modes are:

Sniffer Mode: Reads network packets and displays them in a continuous stream on the console (like when you used -v).

Packet Logger Mode: Records the packets to the disk in a format like PCAP (like when you used -l .).

Network Intrusion Detection System (NIDS) Mode: The most complex mode. It analyzes network traffic against a specific set of rules and takes action (like alert) when a match is found (like when you used -c snort.conf).

2. Which mode is most often used, and why?

NIDS Mode is the most often used.

Why: Because it provides active security. While sniffer and logger modes just show or record data, NIDS mode actually "inspects" the data for malicious patterns and alerts administrators in real-time so they can respond to threats before damage is done.

3. What are some categories of rules that Snort can use?

Based on the icmp.rules you explored and the /etc/snort/rules/ directory, common categories include:

ICMP Rules: For pings and network errors.

Exploit Rules: For known hacking attempts.

Dos Rules: For Denial of Service attacks.

SQL Injection Rules: For database attacks.

Malware-Cnc Rules: For detecting infected machines talking to hackers.

Key Term Quiz

You tell Snort specifically what to look for in the **rules** file.

When you launch Snort, you tell it to read from a **configuration** file.

Through Wireshark, you can view and analyze the **log** file.